

A Physiologically Interpretable Multimodal Model for Joint Sleep Staging and Respiratory-Event Detection in Subacute Ischemic Stroke

MD WAHIDUZZAMAN SUVA¹, ESM-E MOULA CHOWDHURY ABHA¹, MD IMTIAJ ALAM SAJIN¹,
AND MD IFTEKHARUL MOBIN¹

¹Department of Computer Science and Engineering, American International University-Bangladesh, Dhaka 1229, Bangladesh

Corresponding author: Md Iftekharul Mobin (e-mail: iftekhar.mobin@aiub.edu).

The authors received no specific funding for this work.

ABSTRACT Sleep-disordered breathing is common after an ischemic stroke and predicts worse recovery, yet automatic sleep analysis reads only the electroencephalogram and reports only sleep stages. The polysomnogram records far more, and here the disease is visible in its cardiorespiratory channels: the median patient has a scored respiratory event in 13% of epochs. We present MM-Net, a two-stream multi-task model reading fourteen neural and cardiorespiratory channels, which from one forward pass gives the sleep stage of every 30-second epoch and a per-epoch respiratory-event label. On iSLEEPS, a stroke-specific polysomnography corpus (99 patients, 92,560 epochs), under patient-independent ten-fold cross-validation, MM-Net stages sleep at 0.739 accuracy (Cohen's κ 0.634) and flags the epochs carrying a respiratory event at 0.782 AUC, an epoch-level and not event-level read-out, with 2.76 M parameters. Cardiorespiratory input does not raise staging accuracy; it adds the second output at no cost to the first. Against matched single-task models, identical but for the loss, it leads on both tasks, and against the closest prior joint system it leads in every fold on every metric. We benchmark thirteen architectures retrained here and three applied zero-shot from healthy sleep; every retrained baseline lands between 0.613 and 0.720, and the five reimplementations publishing an accuracy are recovered to within 0.04, pointing at the difficulty of the cohort. Repeating the architecture search inside the training data costs 0.006 accuracy. On healthy sleep the same architecture reaches 0.856, losing 0.115 crossing into stroke against 0.140 to 0.148 for those baselines. Code, features and checkpoints are released.

INDEX TERMS Cardiorespiratory signals, feature fusion, ischemic stroke, multi-task learning, multimodal learning, polysomnography, sleep apnea, sleep staging.

I. INTRODUCTION

SLEEP is scored in 30-second epochs into five stages, Wake (W), N1, N2, N3, and rapid-eye-movement (REM), following the American Academy of Sleep Medicine (AASM) rules [1]. The overwhelming majority of automated methods read a single electroencephalogram (EEG) channel and ask one question: how accurately can the five stages be classified? Two decades of work on that question have moved from convolutional and recurrent encoders [2]–[4] through attention and sequence models [5]–[7] to large self-supervised EEG foundation models [8]–[11]. On healthy sleepers these methods reach roughly 85% accuracy, and recent reviews argue the field is near a ceiling set by inter-scorer disagree-

ment [12], [13].

A clinical polysomnogram (PSG), however, is a recording of the whole sleeping body. It carries the EEG, eye movements, muscle tone, the electrocardiogram (ECG), nasal and oral airflow, thoracic and abdominal breathing effort, and peripheral oxygen saturation (SpO₂). These cardiorespiratory channels are where sleep-disordered breathing shows: a paused breath, a chest that keeps straining against a closed airway, a fall in blood oxygen, and the arousal that rescues the breath. A model that reads only the EEG cannot see any of this, and so cannot report on the disorder that most damages a patient's sleep. We read the polysomnogram across both neural and cardiorespiratory physiology,

and we ask a question that conventional staging evaluation does not: what does a model gain, and on which clinical task, from reading the whole recording. We state the answer here, since it governs how the results should be read. The cardiorespiratory channels do not improve sleep staging: the model reaches 0.741 accuracy from the neural stream alone and 0.739 with both, a difference well inside fold-to-fold variation. They provide a second clinical output, respiratory-event detection at 0.782 AUC, from the same forward pass and without reducing staging accuracy. The two physiologies therefore carry complementary information about different clinical questions, and the experiments below are designed to test that separation.

We study this on iSLEEPS [14], released in 2026 and described by its authors as the first Asian and one of the largest stroke-specific sleep databases. The cohort is well matched to the question. Sleep-disordered breathing is common in the weeks after a stroke and is tied to worse recovery [15], [16], and in this corpus the median patient carries a scored respiratory event in 13% of epochs, with 79 of 99 patients above 5% (Fig. 1). The cardiorespiratory signal is therefore abundant and clinically meaningful in exactly the population where breathing matters most. The published staging baselines on this corpus reach 74.7% accuracy for an LSTM, about ten points below the healthy state of the art, and a concurrent study reports that deep networks attend to non-physiological regions on this cohort [17], so the setting also stresses every staging architecture we can bring to it.

We present a two-stream, multi-task model that treats staging and respiratory-event detection as one joint problem, and we place it inside a broad benchmark on the same folds. Our contributions are:

- 1) We introduce MM-Net, a two-stream multi-task model that stages sleep at 0.739 accuracy (κ 0.634) and, from the same forward pass, detects respiratory events at 0.782 AUC, with a direct cardiorespiratory path to the respiratory head so a direct cardiorespiratory path to the respiratory head avoids attenuation of task-specific information during fusion; removing that path costs 0.006 respiratory AUC and leaves staging unchanged (Section V). Against matched single-task models differing only in the surviving loss term, the joint objective is ahead on both outputs, by 0.006 staging accuracy in nine folds of ten and 0.007 respiratory AUC in seven, so the second task is obtained without degrading performance on the first (Section VI-H). The ablation identifies respiratory effort, not oxygen saturation, as the channel the read-out depends on (Section VI-G).
- 2) We benchmark the model against seventeen baseline configurations, those we ran ourselves on the same patient-independent folds as the proposed model, covering convolutional, recurrent, attentional and fully convolutional families, and reimplement six published architectures (U-Time, TinySleepNet, Sleep-Transformer) so that every comparison runs through one pipeline (Table 4). Among them is the closest

prior joint system, Huttunen et al. [18], ported from its authors' released code, and the only baseline that scores both tasks; the proposed model is ahead of it in every fold on every metric.

- 3) We give converging evidence that the accuracy every method reaches here reflects the difficulty of this cohort. The five reimplementations whose papers report an accuracy recover their published Sleep-EDF figure to within 0.04, every retrained baseline lands between 0.613 and 0.720 here, and the proposed model loses 0.115 accuracy crossing from healthy sleep to stroke against 0.140 to 0.148 for those baselines (Section VI-O). These results provide evidence that the observed performance is not primarily attributable to implementation differences.
- 4) We show the respiratory read-out is physiologically grounded: removing the airflow channel from which the events were scored leaves detection at 0.778 AUC, because the decision rests on respiratory effort above all, together with oxygen desaturation, cardiac variability and cortical arousals, encoded redundantly enough that no single channel is indispensable (Section VI-J).
- 5) We attribute each output to the physiology it depends on, by intervention on the model. A leave-one-out modality-ablation grid separates the two outputs: removing the cardiorespiratory stream lowers respiratory detection and leaves staging untouched (-0.128 AUC, Wilcoxon on ten fold-means, Holm $p = 0.018$), while no neural ablation moves respiratory AUC. That double dissociation is a falsifiable account of what each modality contributes (Table 7).

Together, these results reframe the problem on this cohort: the useful gains arise from integrating neural and cardiorespiratory physiology and attributing each output to the modality that supports it, while deeper single-channel staging does not help here. The remainder of the paper develops the model (Section V) and tests each claim in turn.

II. RELATED WORK

A. SINGLE-CHANNEL DEEP SLEEP STAGING

Automated staging matured on one EEG channel. DeepSleepNet paired convolutional feature extraction with a bidirectional LSTM [2]; SeqSleepNet framed the night as sequence-to-sequence labeling [3]; AttnSleep introduced attention over multi-resolution convolutions [5]; TinySleepNet and U-Time reduced the parameter budget while holding accuracy [4], [7]; and XSleepNet fused raw and spectral views [6]. More recent backbones push this further: SleepTransformer adds interpretability and uncertainty to an attention model [19], L-SeqSleepNet models whole-night cycles [20], ZleepAnlystNet and FlexSleepTransformer explore separated training and flexible channel sets [21], [22], and sleep stagers built on the linear-time state-space backbone of Mamba [23] reach competitive accuracy [24]. These models set the accuracy on healthy corpora but depend on the cortical micro-events

that stroke disturbs, which is why they transfer poorly to patients [17].

B. MULTI-CHANNEL AND FOUNDATION MODELS

Later work reads several PSG channels. U-Sleep stages from arbitrary EEG and EOG derivations across many cohorts [25]. Joint scoring of the two tasks is itself established. Huttenen et al. trained a single network to score sleep stages and respiratory events simultaneously across 877 clinical recordings, comparing signal combinations from pulse oximetry alone up to a full montage [18], and multitask staging-and-apnea models have followed [26]. Those studies work in suspected-apnea and elderly populations; what is untested is whether the arrangement survives a cohort whose neural signal is itself damaged, which is the question this paper asks and which Section VI answers by running their architecture here, on these folds, from the code they released. Foundation models such as LaBraM, CBraMod, EEGPT, and BIOT learn transferable representations from large unlabeled corpora [8]–[11]. SleepFM extends this idea to several PSG modalities [27], while cross-scale foundation models and standardized benchmarks continue to appear [28]–[30], self-supervised pretraining is now systematically evaluated for label efficiency [31], transfer and domain adaptation improve robustness across cohorts [32], and recent surveys and clinical benchmarks catalogue the field [13], [33]. The eye and muscle channels help most with REM and Wake, which a single EEG channel resolves poorly. Reported gains from adding respiratory channels to *staging* are small and cohort-dependent; our benchmark quantifies this directly on a high-burden clinical cohort and separates the staging effect from the respiratory one.

C. DETECTION OF SLEEP-DISORDERED BREATHING

A parallel line of work detects apnea and hypopnea from breathing signals. Airflow and effort shape, oximetry, and ECG-derived heart-rate variability are all informative, and oxygen desaturation is part of the scoring criterion for a hypopnea [1]. These detectors usually run at fine time resolution on the raw breathing waveforms and are trained on that task alone. We instead detect respiratory events on the staging epoch grid, jointly with staging, from a shared representation, so that one model produces both clinical outputs and the two tasks support each other during training.

D. SLEEP AND BREATHING AFTER STROKE

Sleep is both disturbed by stroke and predictive of recovery, and sleep-disordered breathing is common in the subacute phase and linked to poorer functional outcome [15], [16], [34]. Focal injury also changes the micro-events that scoring relies on, the spindles, slow waves, and muscle atonia [35]–[39]; quantitative EEG markers such as the brain symmetry index further track hemispheric injury [40]. A model that reports on breathing as well as stage is therefore of direct clinical value in this population. Sleep staging on ischemic stroke itself is nascent, reported only as baselines on the iSLEEPS

corpus [14], and no prior method addresses staging and respiratory detection jointly, on a stroke population. Rather than set published accuracies side by side, each measured on its own corpus and protocol, we re-run the architectures that matter on these folds (Table 4).

III. CRITICAL GAPS AND LIMITATIONS OF PRIOR WORK

Three gaps run through the literature above, and each maps onto one of our contributions. (i) *Single-task, single-signal design*. Almost every staging model reads one or a few EEG channels and predicts only the stage, while apnea detectors read breathing signals and predict only events. Multimodal foundation models have begun to join them, SleepFM [27] staging sleep and classifying apnea severity from one representation on a general population, but per-epoch respiratory detection alongside staging has not been shown on a stroke cohort, leaving the clinician with two separate tools. Contribution 1 answers this with a single model that reads both physiologies and produces both outputs. (ii) *Evaluation on healthy sleepers*. Staging accuracy is reported almost entirely on healthy corpora such as Sleep-EDF [41], [42], where deep and foundation models reach a ceiling, while performance on focal brain injury is largely unmeasured. Contributions 2 and 3 benchmark strong deep architectures, and classical gradient-boosting pipelines [43]–[45] on a subacute-stroke cohort, show substantial degradation of architectures developed on healthy-sleep cohorts, and provide evidence by reimplementing that this degradation is associated with the cohort rather than with implementation differences. (iii) *Attribution and reliability*. Interpretability is usually post-hoc and unfalsifiable, and predictive uncertainty is seldom modeled [46]. Contributions 4 and 5 answer this with a physiologically grounded, non-circular respiratory read-out and an interventional modality-ablation that attributes each output to the channel whose physiology is closest to it.

IV. DATASET AND PREPROCESSING

iSLEEPS [14] contains overnight PSG from 100 subacute ischemic-stroke patients (mean age 50.5; 77 male, 23 female), scored by experts into the five AASM stages. We exclude one recording after an audit: the data records of SN28 are byte-identical to those of SN15, a duplication consistent with the anonymization and re-numbering the dataset authors describe, and keeping it would place a training subject into the test fold in nine of ten fold assignments, giving $N = 99$. Both tasks use all 99 patients: the twelve recordings that lack at least one cardiorespiratory channel are scored with those channels zero-filled instead of excluded, and we state which N applies to each reported number. The corpus totals 92,560 epochs. The stage distribution (Table 1) is strongly imbalanced, with N2 dominant and N1 and N3 scarce, which is the clinical norm.

a: Channels we read, and channels we do not

A raw recording holds 22 traces. We read the 14 that carry sleep-stage or breathing information. The *neural stream* takes

TABLE 1. Stage distribution and respiratory-event prevalence in the corpus we model ($N = 99$). Respiratory events occupy 16.0% of all epochs, and the per-patient burden ranges from 0 to 60.5%, with 79 of 99 patients above 5% (Fig. 1).

Stage	W	N1	N2	N3	R
Share (%)	26.6	10.2	42.3	8.9	12.1

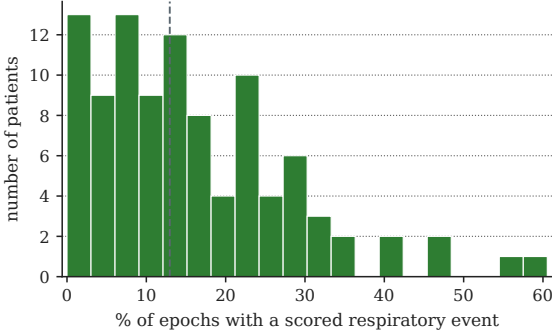


FIGURE 1. Sleep-disordered-breathing burden across the cohort: patients by the fraction of epochs carrying a scored respiratory event, over the 99 with respiratory labels. The dashed line marks the median, 13%.

the four EEG derivations (C4:M1, C3:M2, O2:M1, O1:M2), the two EOG channels (E1:M2, E2:M2), and the submental EMG. The *cardiorespiratory stream* takes the ECG, nasal and oral airflow, thoracic effort, abdominal effort, the summed effort trace, SpO₂, and pulse. We leave out eight traces for stated reasons: periodic limb movement records a different disorder; snore is scored from the same airway sound as the respiratory events and would make the second task circular; plethysmography is already summarized by the SpO₂ and pulse channels we keep; body position is coarse and slow; and the ambient-light and battery traces are device house-keeping, not physiology. All neural channels are resampled to 100 Hz for the engineered descriptors, and independently to 200 Hz for the pretrained encoder as described below; the cardiorespiratory channels go to a common 25 Hz grid by polyphase resampling, which applies the anti-aliasing filter the rate change requires; SpO₂, recorded natively at 4 Hz, is carried to this grid by the same operation, which adds samples and no information, so its effective resolution remains that of the 4 Hz oximeter. Signals are read in physical units, which matters because SpO₂ near 90%, pulse near 78 bpm, and microvolt EEG live on very different scales. Per-epoch respiratory labels are the corpus's scored events mapped onto the 30 s grid, and an epoch is positive when it overlaps an event.

V. PROPOSED METHOD

A. PROBLEM FORMULATION

A night is a sequence of T epochs. For epoch t we form a neural feature vector $\mathbf{f}_t^{\text{eeg}} \in \mathbb{R}^{388}$, formed by concatenating the 188 engineered descriptors below with a 200-dimensional embedding from LaBraM, a frozen pretrained EEG encoder [8] (Section VI-P), and a raw cardiorespiratory

tensor $\mathbf{x}_t^{\text{car}} \in \mathbb{R}^{7 \times 750}$, the seven channels at 25 Hz. The 14 engineered cardiorespiratory descriptors defined below are used by the baseline configuration against which the learned encoder is measured.

The pretrained encoder is fed on its own terms. LaBraM was pretrained at 200 Hz and fixes its input length at 3000 samples, so the EEG is resampled from our 100 Hz working rate to 200 Hz before it reaches the encoder. Feeding the 100 Hz signal directly would have been silent but wrong: 3000 samples then span 30 seconds of recording while the encoder reads them as 15, compressing time by a factor of two, so every rhythm would present at twice its frequency and a 12 Hz spindle would arrive as 24 Hz. At 200 Hz a 30 s epoch no longer fits the positional embedding in one piece, so each epoch is split into two 15 s sub-windows of 3000 samples, encoded separately, and mean-pooled; the two embeddings were checked to differ before caching, so the pooling carries information and does not merely average duplicates. The encoder also requires canonical 10–20 names and rejects mastoid references, so C4:M1, C3:M2, O2:M1 and O1:M2 are presented as C4, C3, O2 and O1, and the ocular and chin channels are dropped for having no canonical EEG name. Amplitudes are scaled to $\mu\text{V}/100$. The encoder therefore sees four EEG channels, strictly fewer signals than the engineered descriptors read. We learn a subject-independent map

$$f_{\theta} : (\mathbf{f}_{1:T}^{\text{eeg}}, \mathbf{x}_{1:T}^{\text{car}}) \mapsto (\hat{\mathbf{y}}_{1:T}^{\text{stg}}, \hat{\mathbf{y}}_{1:T}^{\text{apn}}), \quad (1)$$

where $\Delta^4 = \{p \in \mathbb{R}^5 : p_k \geq 0, \sum_k p_k = 1\}$ is the four-simplex, $\hat{\mathbf{y}}_t^{\text{stg}} \in \Delta^4$ is a distribution over the five stages and $\hat{\mathbf{y}}_t^{\text{apn}} \in [0, 1]$ scores whether epoch t contains a respiratory event (Fig. 2). We write $\hat{\mathbf{y}}_t^{\text{apn}}$ as a score and not as a probability for the reason given in Section V: the positive weighting in its loss displaces the optimum away from the conditional event rate.

B. PHYSIOLOGICAL FEATURE REPRESENTATIONS

a: Neural features (188)

Per channel we take absolute and relative band powers from the Welch spectrum $P(f)$ in delta (0.5–4), theta (4–8), alpha (8–13), sigma (11–16), and beta (16–30) Hz; the relative power in band b is

$$\text{RP}_b = \frac{\int_{f \in b} P(f) df}{\int_{0.5}^{30} P(f) df}. \quad (2)$$

The bands are the conventional sleep bands and alpha and sigma overlap between 11 and 13 Hz, so the five relative powers are not a partition and do not sum to one. We keep the conventional definitions because both bands name physiology a scorer reads, and the model treats the five as correlated descriptors whose sum is unconstrained. We add spectral entropy $H = -\sum_i p_i \log p_i$ with p_i the normalized spectrum, the 95% spectral-edge frequency, the three Hjorth descriptors [47], and time-domain statistics. Sleep spindles are detected on the sigma-band analytic envelope, the modulus of the analytic signal built from the Hilbert transform $\mathcal{H}$,

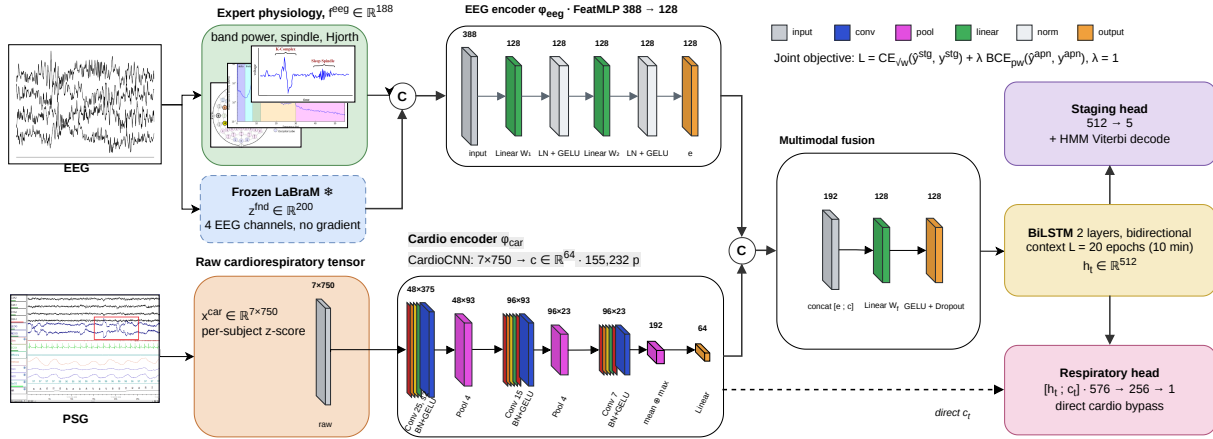


FIGURE 2. The two-stream, multi-task model. Layer shapes above each block, operation below; block height encodes units for the multilayer perceptron and temporal length for the convolutional stack. Dashed outlines mark frozen parameters. The junction marked C is the concatenation used for every reported result, and the fusion panel is the single linear layer that follows it.

$e(t) = |x_\sigma(t) + i\mathcal{H}\{x_\sigma\}(t)| = \sqrt{x_\sigma(t)^2 + \mathcal{H}\{x_\sigma\}(t)^2}$, with density counted above a per-epoch threshold. No duration, separation or morphology criterion is applied, so this counts crossings of a sigma-band envelope and is a proxy for spindle density and not a validated spindle detector; the same holds for the slow-wave descriptors, which summarise amplitude and detect nothing. Ocular-movement energy on the EOG and tonic EMG level complete the set.

b: Cardiorespiratory features (14)

From the seven cardiorespiratory channels we compute SpO₂ mean, minimum, variability, and desaturation depth (median minus minimum); pulse mean and variability; ECG amplitude and line-length; airflow amplitude and line-length; thoracic, abdominal, and summed effort amplitudes; and signed thoraco-abdominal synchrony, whose loss is the paradoxical motion of chest against abdomen that marks an obstructed breath. The correlation is kept signed, so synchronous and paradoxical motion sit at opposite ends of the range; a value near +1 is synchronous motion and a negative value is paradox,

$$\text{sync}_t = \text{corr}(x_t^{\text{tho}}, x_t^{\text{abd}}). \quad (3)$$

Both feature sets are standardized per patient, which removes person-to-person offsets so the model generalizes across people. Appendix A lists all 188 neural descriptors with the parameters used to compute them.

c: The learned cardiorespiratory encoder

The evaluated model does not read the 14 descriptors. Its cardiorespiratory branch is a three-layer convolutional encoder over the raw 7×750 tensor, with wide kernels and aggressive downsampling because respiratory events are slow: at 25 Hz the first kernel of 25 samples spans one second, and a hypopnea unfolds over 10 to 30. Channel widths are 48, 96 and 96, each followed by batch normalisation and GELU, with max-pooling by four after the first two. Mean and max pooling

over time are concatenated and projected by a linear layer, LayerNorm, GELU and dropout to the same 64 dimensions the feature MLP produced, so the encoder is a drop-in replacement and every other part of the network is unchanged. The branch holds 155,232 trainable parameters, of which the projection LayerNorm supplies 128. Section VI-P quantifies the effect of this substitution, which is the largest of any representation change we measure, and Section VI-B shows that the effect belongs to the change of representation: random encoders of comparable capacity over the same raw signal recover much of it. This encoder is used because it is small, trained end-to-end with the rest of the network, and adds no external dependency. We therefore attribute the observed improvement to the representation rather than to the specific encoder architecture.

C. TWO-STREAM ENCODERS

The neural stream is a two-layer perceptron with LayerNorm; the cardiorespiratory stream is the convolutional encoder described above, which emits a vector of the same width the perceptron it replaced produced,

$$e_t = \text{GELU}(\text{LN}(W_2 \text{GELU}(\text{LN}(W_1 f_t^{\text{eeg}})))) \in \mathbb{R}^{128}, \quad (4)$$

$$c_t = \text{CardioCNN}(x_t^{\text{car}}) \in \mathbb{R}^{64}. \quad (5)$$

LayerNorm is used in place of BatchNorm because the recurrent windows are short and mix patients within a batch, and the cardiorespiratory stream is deliberately narrower because the branch compresses a 7×750 tensor into the same 64 dimensions the feature encoder it replaced produced, which keeps the substitution the only variable.

D. FUSION

The two embeddings are concatenated and projected back to the stream width,

$$z_t = \text{GELU}(W_f [e_t; c_t]) \in \mathbb{R}^D, \quad D = 128. \quad (6)$$

This is the simplest fusion available, and it is what every number in this paper was produced with. We additionally evaluated an attention-based fusion, in which the two embeddings are treated as tokens carrying learned modality-type embeddings, refined by four-head scaled dot-product attention so each modality can read the other, then projected. On the same ten folds, in the feature-only configuration the comparison was run in, attention was indistinguishable from concatenation on staging (+0.005 accuracy, $p = 0.49$) and slightly worse on the respiratory head (-0.014 AUC, $p = 0.06$). Since the attention-based fusion provides no measurable improvement in this setting, we report and release the simpler one, and the gain from the second stream is therefore attributable to the added modality.

E. TEMPORAL DECODER AND TWO HEADS

Sleep is sequential and respiratory events cluster, so the fused vectors of an $L = 20$ epoch window (10 minutes) are read by a two-layer bidirectional LSTM. Training draws windows at a stride of $L/2$, so an interior epoch is seen twice per pass and an epoch in the first or last ten of a recording once; we apply no correction for that, and the first and last window of each night are zero-padded to L under a mask that removes the padding from the loss. Inference uses disjoint windows: a recording is padded to a multiple of L , split into consecutive blocks, and the padding discarded, so every epoch is scored exactly once and no stitching rule is needed. Hidden states are not carried between windows in either case. Each direction runs the standard recurrence

$$\begin{aligned} i_t &= \sigma(W_i[z_t, h_{t-1}]), & f_t &= \sigma(W_f[z_t, h_{t-1}]), \\ o_t &= \sigma(W_o[z_t, h_{t-1}]), & g_t &= \tanh(W_g[z_t, h_{t-1}]), \\ s_t &= f_t \odot s_{t-1} + i_t \odot g_t, & h_t &= o_t \odot \tanh(s_t), \end{aligned} \quad (7)$$

and the forward and backward states of the 256-unit layers are concatenated into $h_t \in \mathbb{R}^{512}$. The staging head is a linear map $\hat{y}_t^{\text{stg}} = \text{softmax}(W_s h_t)$. The respiratory head reads h_t together with a direct copy of the per-epoch cardiorespiratory embedding c_t ,

$$\hat{y}_t^{\text{apn}} = \sigma(W_2 \text{GELU}(W_1[h_t; c_t])). \quad (8)$$

The connection exists so that the respiratory decision does not have to survive a fusion trained mostly for staging, which would weaken the mechanical signature of an obstructed breath along the way. Section VI-G identifies the channel that signature lives in, and it is respiratory effort that names the event. We isolate the connection by removing it and holding everything else fixed, on the same folds as every other figure here. Respiratory AUC falls from 0.782 to 0.776 (Wilcoxon on ten fold-means, $p = 0.049$) and average precision from 0.419 to 0.411 ($p = 0.28$), while staging is untouched at 0.739 accuracy ($p = 0.92$) and κ 0.634 ($p = 0.92$). The gain is small, about a sixth of a fold standard deviation, and the p -value is marginal, so it does not support an argument on its own. Its relevance lies in where the effect appears: the

connection feeds the respiratory head and nothing else, and the respiratory head is the only output that changes.

Capacity is not what produces that gain. The head reads 2 hidden+ d_{card} inputs whether or not the bypass is used, since the ablation zeroes the cardiorespiratory block instead of deleting it, so both arms hold the same 2,764,774 parameters. A third configuration provides a stricter control by feeding the head the same embeddings permuted across the epochs of the window, which keeps capacity, statistics and gradient path and destroys only the alignment to the epoch. It reaches 0.774 AUC, at or just below the zeroed arm's 0.776, indicating that a misaligned cardiorespiratory signal confers no advantage over its absence. On this evidence the bypass is a supporting design choice.

The trained model holds 2.76 million parameters. The pretrained encoder that supplies the 200-dimensional neural embedding is run once offline and is not trained here, so it adds no trainable parameters to this count.

F. TRAINING AND DECODING

The two tasks are trained together with equal weight, $\mathcal{L} = \mathcal{L}^{\text{stg}} + \lambda \mathcal{L}^{\text{apn}}$ with $\lambda = 1$, combining a class-weighted staging cross-entropy and a positive-weighted respiratory cross-entropy,

$$\mathcal{L}^{\text{stg}} = - \sum_t \sum_k w_k \hat{y}_{t,k}^{\text{stg}} \log \hat{y}_{t,k}^{\text{stg}}, \quad (9)$$

$$\mathcal{L}^{\text{apn}} = - \sum_t [\pi a_t \log \hat{y}_t^{\text{apn}} + (1 - a_t) \log(1 - \hat{y}_t^{\text{apn}})]. \quad (10)$$

The staging weights use the square root of inverse class frequency, $w_k \propto \sqrt{N/(Kn_k)}$ for $K = 5$ classes with n_k examples, which corrects the scarce N1 and N3 stages without giving up overall accuracy, and the respiratory term is positive-weighted by $\pi = n_-/n_+$ to match the 16% event rate, which on this corpus is $\pi \approx 5.3$.

Both weightings move the optimum, and we state where. Minimising the weighted binary term against a true conditional rate p is minimised not at p but at $q^* = \pi p / [\pi p + (1 - p)]$, so an epoch carrying a true event probability of 0.157 is driven towards 0.5; the staging weights displace the softmax by the same mechanism, to $q_k^* \propto w_k p_k$. Neither head therefore emits a calibrated posterior, and a threshold of 0.5 on the respiratory output is not a fifty-per-cent event probability. Both displacements are monotone, so ranking is untouched and the AUC and average precision we report are unaffected; what they do affect is any reading of the numbers as probabilities, which is why Section VI treats the outputs as scores and measures calibration separately.

At inference the staging posteriors are smoothed by a Viterbi pass whose transition matrix A^H and prior are counted from the training labels [48],

$$\delta_t(j) = \max_i [\delta_{t-1}(i) + \log A_{ij}^H] + \log \hat{y}_t^{\text{stg}}(j), \quad (11)$$

recovered by back-tracking, which restores the natural persistence of sleep.

This is a transition-regularised discriminative decoder and not a generative hidden Markov model. A generative emission term is $\log P(\mathbf{x}_t | s_t = j)$, whereas $\hat{y}_t^{\text{stg}}(j)$ is a class-weighted discriminative score computed from a bidirectional window; inserting it in the emission slot applies the stage prior a second time, once through A^H and once through the score itself. Dividing by the prior would not repair this, because the class weights and the recurrent context have already displaced the score from any posterior it might have been. The pass supplies a transition prior on the output sequence, which is the property we claim for it. The respiratory head is left unsmoothed. Optimization uses AdamW with cosine annealing and early stopping on a held-out set of patients.

VI. EXPERIMENTS

A. PROTOCOL AND METRICS

All results use ten-fold, patient-independent cross-validation, so no patient appears in both training and test, and within each training split a disjoint set of patients is held out for early stopping. Staging is scored by accuracy, macro-F1 $\frac{1}{K} \sum_k F1_k$, and Cohen's κ . The respiratory task is scored by the area under the ROC curve (AUC) and average precision (AP), which suits the 16% event rate.

Every number we report is a mean over the ten folds with its standard deviation. That standard deviation is a spread and not a confidence interval, and the distinction matters here because the folds share training patients and are therefore dependent. The unit that is close to independent is the patient, so we also resample patients with replacement, 10,000 times, rebuilding the pooled epoch set and recomputing every metric. That gives 0.734 accuracy with a 95% interval of [0.717, 0.750], κ 0.628 [0.604, 0.651], respiratory AUC 0.778 [0.752, 0.802] and average precision 0.419 [0.355, 0.480]. These condition on the fitted procedure: they carry the variation in which patients were recorded and not retraining variability, and they say nothing about the selection optimism measured below. The folds are fixed across all configurations so that comparisons are paired. Fixing them has a cost that we quantify below: the reported configuration was chosen by comparing forty-nine candidates on these same ten folds, which is the standard route by which an in-domain figure acquires selection optimism. We therefore measured how much it acquired, by repeating the selection with the test patients held out from model selection (Section VI-B): five outer patient-independent folds, all forty-nine candidates compared on three inner folds within each outer training set, and the winner scored once on outer patients that took no part in choosing it. That procedure reaches 0.734 ± 0.009 accuracy against the 0.739 reported here, so the optimism is 0.006, about a third of the fold-to-fold standard deviation of 0.019 and small enough that it changes no comparison in this paper. The external corpora in Section VI-S took no part in selection either, so the zero-shot figures there were never exposed to it. The proposed model, its ablations and the six reimplemented literature models all use the full ten; the earlier retrained baselines were run at five, which Table 4 states per row. Because performance

estimates may vary across random seeds, we retrained the headline configuration under three seeds on the same folds. Staging accuracy varies by 0.005 across seeds against 0.020 across folds, and respiratory AUC by 0.001 against 0.038: fold-to-fold spread is four times the seed spread for staging and thirty times it for respiratory detection. The uncertainty in these results is therefore dominated by which patients fall in which fold, which is a property of a cohort this size. Every headline figure in this paper is the mean over all thirty fold-values, ten folds by three seeds. Paired tests, however, are computed on the ten fold-means: the three seeds within a fold train and test on the same patients, so they constitute repeated training runs on one patient partition, and treating them as thirty would triple the effective sample and shrink every p . One consequence follows directly. With ten paired folds the smallest two-sided Wilcoxon p is $2/2^{10} = 0.002$, reached when every fold moves the same way, so after Holm correction across a family of nine the floor is 0.018. Several effects below sit at that floor and differ in size by more than an order of magnitude; the test cannot separate them, and the effect sizes are reported beside every p for that reason. The three seed means for respiratory AUC are 0.781, 0.783 and 0.781, so 0.782 is stable to the third decimal under reseeding.

B. SELECTION INSIDE THE TRAINING DATA

The figures above are produced by a configuration chosen on the folds that score it. To find out what that contributes we ran the whole selection again with the test patients held out from model selection: five outer patient-independent folds; inside each outer training set, three inner folds; all forty-nine candidates of the original search compared on those inner folds alone; and the inner winner retrained on the full outer training set and scored once on the outer patients. No outer patient influences the choice of the model that scores them, so the resulting number estimates the whole procedure, search included.

The nested procedure yields a comparable estimate, reaching 0.7336 ± 0.0089 accuracy across the five outer folds against 0.739 for the reported model. The estimated selection optimism is therefore 0.006 accuracy, about a third of the fold-to-fold standard deviation and smaller than the performance differences underlying the principal comparisons. It is also close to the 0.004 suggested by the spread of the top ten candidates, so the bound available without a nested loop was of the right order.

The design the inner loop arrived at is informative in its own right. Every outer fold kept the neural side of the published model unchanged, the engineered descriptors with the frozen encoder, the BiLSTM, and the same width and regularisation, providing consistent support for the neural half of the architecture across all five outer folds. Where the folds differed from the published model was the cardiorespiratory encoder: four chose a randomly initialised time-series encoder over the raw signal and the fifth a random projection of the fourteen engineered descriptors, and none chose the learned convolutional encoder.

TABLE 2. Training configuration of the proposed model, shared across all folds and every ablation. Each reimplemented baseline is trained at its own published settings instead.

Setting	Value
Framework / hardware	PyTorch, NVIDIA RTX 2060 (6 GB)
Optimizer	AdamW
Learning rate	3×10^{-4}
Weight decay	1×10^{-4}
LR schedule	cosine annealing
Sequence length L	20 epochs (10 min)
Batch size	32
Max epochs / patience	45 / 8
Gradient clipping	5.0
Staging class weights	$\sqrt{\text{inverse-frequency}}$
Respiratory pos. weight	$\pi = n_- / n_+$
Loss balance λ	1.0
Staging post-processing	HMM Viterbi smoothing
Cross-validation	10-fold, patient-independent
Random seeds	42, 1, 7

That is consistent with what Section VI-P already reports from the other direction, where a random expansion of the same fourteen numbers recovers a large part of the gap to the learned encoder. Read together, the two experiments locate the effect: what matters is that the cardiorespiratory stream is given a high-capacity representation of the raw signal rather than fourteen summary statistics, and several encoders supply that equally well at this cohort size. These results therefore attribute the observed effect to the change in representation rather than to the specific encoder architecture. The encoder implementing it is interchangeable at this cohort size, and the accuracy is unaffected either way, since every configuration the search selected lands within 0.006 of the reported one.

Two limits. Five outer folds are fewer than the ten used everywhere else, so this estimate is noisier than the figures it is compared against; and it uses a single seed, because five outer folds by forty-nine candidates by three inner folds is already seven hundred and thirty-five trainings.

C. IMPLEMENTATION AND HYPERPARAMETERS

The model is implemented in PyTorch and trained on a single NVIDIA RTX 2060 GPU (6 GB) under Windows. Every run is seeded, so each is individually reproducible; headline figures average three seeds (42, 1, 7) over the ten folds, giving the thirty fold-values used throughout. Where a result is derived from stored predictions without retraining, it comes from the seed-42 run, and the text says so at that point. Table 2 lists the training configuration, which is identical across all folds and every ablation, so that reported differences come from the model. The six reimplemented baselines are trained at the optimiser, batch size and schedule of their own papers, since imposing our settings on them would understate architectures tuned for other choices.

TABLE 3. Reimplementation check for the models marked [†] in Table 4: ten-fold subject-independent cross-validation on the Sleep-EDF Cassette subjects 0–20, 40 recordings, single Fpz-Cz derivation. Every row compares accuracy except [§]U-Time, whose paper reports macro-F1 and no accuracy for Sleep-EDF; that row therefore compares macro-F1 to macro-F1, and our U-Time reaches 0.791 ± 0.062 accuracy, a quantity its paper does not state. Each published figure follows its own paper's Sleep-EDF protocol, which differs in subset and split.

Model	Published	Ours	Δ
U-Time [7] [§]	0.79	0.660 ± 0.054	−0.130
TinySleepNet [4]	0.854	0.824 ± 0.045	−0.030
SleepTransformer [19]	0.814	0.823 ± 0.033	+0.009
1D-ResNet-SE-LSTM [49]	0.864	0.832 ± 0.036	−0.032
Micro SleepNet [50]	0.828	0.800 ± 0.041	−0.028
AT-BiLSTM [51]	0.838	0.835 ± 0.046	−0.003

D. COMPARED METHODS

We compare the model against deep networks evaluated on the same folds: convolutional and recurrent networks trained from scratch (DeepSleepNet, AttnSleep, a four-channel CNN–BiLSTM), six further published architectures we reimplemented for this comparison (U-Time, a fully convolutional encoder–decoder; TinySleepNet, a compact convolutional–recurrent model; and SleepTransformer, a purely attentional one), a network transferred from Sleep-EDF, and a raw-signal convolutional network on all 14 channels, alongside the published single-channel baselines for the corpus [14]. All numbers are patient-independent, ten-fold except for the earlier retrained baselines at five, at the hidden-Markov operating point where a model provides one.

E. STAGING BENCHMARK AND COMPARISON

The row marked [‡] is quoted from the corpus paper and was not run here. It uses that paper's protocol, an 80/10/10 patient-wise split over $N = 100$ that includes the duplicated pair we exclude (Section IV), and its own two-channel input, and that paper's repository releases preprocessing only, so it can be neither reproduced nor read as a same-protocol comparison. Each reimplementation is checked against the figure its own paper publishes on Sleep-EDF (Table 3); those figures follow each paper's own protocol, which differs in subset and split, so they are consistency checks. Our own split uses one subject more than the conventional Sleep-EDF-20 set, and one of the published figures is measured on SHHS, so those two checks are weaker than the rest. A third is weaker still. U-Time publishes macro-F1 and not accuracy, so its row is the only one compared in macro-F1, and there our reimplementation sits 0.130 below the published figure. Macro-F1 is governed by N1, the stage whose prevalence moves most between Sleep-EDF subsets, and the published figure comes from the 39-recording subset under twenty-fold cross-validation against our 40 recordings under ten. The discrepancy cannot be attributed reliably to reimplementation error, because the evaluation protocols differ substantially, and we therefore do not treat this row as a direct quantitative comparison. Table 4 is organised in three blocks: architectures retrained on this cohort, architectures trained on healthy sleep and applied

TABLE 4. Sleep staging on iSLEEPS, patient-independent, against every architecture we evaluated. Cells are mean $\pm$ SD across folds where per-fold values were retained. [†] marks a model we reimplemented and checked against its published Sleep-EDF figure (Table 3), [‡] one quoted from the corpus paper on its own protocol, “feat” engineered features. * marks the one architecture ported from its authors’ released code rather than reimplemented from a paper; it carries no row in Table 3 because that paper reports no Sleep-EDF figure to check against, and it is the only baseline here that scores both tasks. Bold marks the best value among rows run under our protocol.

Model	Input	Params ↓	Healthy	iSLEEPS (stroke)		
			Acc	Acc ↑	Macro-F1 ↑	κ ↑
<i>Deep networks, retrained on iSLEEPS</i>						
LSTM, our reimplementation [†]	1 EEG + EOG	1.16 M	—	0.720 ± 0.032	0.626 ± 0.028	0.599 ± 0.043
CNN-ResNet18 [17] [‡]	1 EEG + EOG	~11 M	—	0.617	0.544	0.480
DeepSleepNet [2]	1 EEG	~21 M	—	0.615 ± 0.026	0.567 ± 0.031	0.483 ± 0.038
U-Time [7] [†]	1 EEG	1.07 M	0.791 ± 0.062	0.646 ± 0.064	0.524 ± 0.057	0.506 ± 0.082
AttnSleep [5]	1 EEG	~0.5 M	—	0.716 ± 0.042	0.600 ± 0.043	0.593 ± 0.058
TinySleepNet [4] [†]	1 EEG	1.45 M	0.824 ± 0.045	0.676 ± 0.050	0.587 ± 0.034	0.548 ± 0.062
SleepTransformer [19] [†]	1 EEG	2.79 M	0.823 ± 0.033	0.683 ± 0.026	0.602 ± 0.030	0.555 ± 0.032
1D-ResNet-SE-LSTM [49] [†]	1 EEG	0.65 M	0.832 ± 0.036	0.708 ± 0.038	0.631 ± 0.033	0.592 ± 0.050
Micro SleepNet [50] [†]	1 EEG	0.03 M	0.800 ± 0.041	0.647 ± 0.049	0.558 ± 0.028	0.510 ± 0.055
AT-BiLSTM [51] [†]	1 EEG	0.92 M	0.835 ± 0.046	0.692 ± 0.040	0.612 ± 0.030	0.570 ± 0.051
CNN + BiLSTM	4 EEG	~0.6 M	—	0.613 ± 0.040	0.565 ± 0.034	0.485 ± 0.048
Sleep-EDF transfer	4 EEG	~0.6 M	—	0.623 ± 0.020	0.582	0.488
Raw multimodal CNN	14 ch	0.95 M	—	0.654 ± 0.033	0.610 ± 0.022	0.544 ± 0.037
Huttunen et al. [18]*	4 ch	3.80 M	—	0.599 ± 0.060	0.495 ± 0.067	0.440 ± 0.078
<i>Trained on healthy sleep, applied zero-shot</i>						
CNN	4 EEG	0.49 M	0.806 ± 0.001	0.307 ± 0.016	0.223 ± 0.013	0.102 ± 0.010
CNN + BiLSTM (CRNN)	4 EEG	1.15 M	0.828 ± 0.005	0.312 ± 0.012	0.253 ± 0.006	0.127 ± 0.022
StagingSeqNet	4 EEG	1.08 M	0.855 ± 0.001	0.306 ± 0.009	0.230 ± 0.014	0.091 ± 0.015
<i>Proposed feature-based multi-task model</i>						
MM-Net (neural only)	7 ch feat	2.62 M	—	0.741 ± 0.017	0.668 ± 0.020	0.635 ± 0.022
MM-Net (neural + cardio)	7 ch feat + raw	2.76 M	—	0.739 ± 0.020	0.670 ± 0.020	0.634 ± 0.024
<i>Same model, healthy sleep</i>						
MM-Net (neural only)	2 EEG feat	2.62 M	0.856 ± 0.027	—	—	—

to it zero-shot, and the proposed model, with a final row cross-validating the proposed architecture on healthy sleep as a control on the cohort argument; its macro-F1 and κ are healthy-sleep figures and so are left out of the stroke columns, and quoted in the text instead. Dispersion is reported wherever it is defined: five folds for the earlier retrained baselines; thirty fold-values, ten folds by three seeds, for AttnSleep, the three reimplemented literature models and the proposed rows. CNN-ResNet18 is quoted from the corpus paper and was not re-run here, so its protocol is that paper’s and it carries no interval. That paper’s LSTM appears only as our reimplementation of it on the folds used throughout, because its published figure was never measured on these folds. In the zero-shot block the interval is across pretraining seeds and not folds, because that evaluation is a single pass of a finished checkpoint over all 99 patients. The two best healthy scores, 0.856 and 0.855, are separated by less than either interval. Sleep-EDF carries none of the cardiorespiratory channels this model reads, its Cassette recordings providing an oronasal airflow trace at 1 Hz but no ECG, effort belts or oximetry and no scored respiratory events, so every healthy-sleep figure exercises the neural branch alone and the comparison row there is MM-Net (neural only).

Table 4 shows a clear pattern: every deep network, whether trained from scratch, transferred from healthy sleep, or applied to the raw multichannel signal, stays between 0.61

and 0.72 accuracy, 12 to 20 points below the same architectures’ published accuracy on healthy sleepers. The feature-based model reaches 0.739, above every deep architecture we retrained here, which matches the concurrent finding that depth alone does not help on this cohort [17]: with under a hundred patients a convolutional network cannot recover the physiological structure the engineered descriptors encode explicitly, which suggests that those descriptors supply useful inductive structure at this sample size. The corpus paper’s own recurrent baseline, reimplemented here and run on the same folds, reaches 0.720 $\pm$ 0.032 over thirty fold-values. Staging performance remains within a narrow range across the evaluated methods: nothing we have run on this corpus exceeds 0.75. Section VI-N reports evidence consistent with this observation, in that the staging learning curve has flattened. We therefore do not position the model as a staging leader. Its principal distinction is the additional respiratory output. Of the models that stage competitively here it is the only one that also detects respiratory events; the raw multimodal CNN produces both outputs but stages at 0.654.

The comparison that speaks to this directly is against Huttunen et al., the closest prior system: one network scoring sleep stages and respiratory events at once, which makes it the only baseline in Table 4 that answers the same two questions this model does. We ported the architecture from the authors’ released code, so the encoder blocks, the squeeze-

and-excitation ratios, the atrous pyramid at the bottleneck and the shape of the two segment classifiers are theirs. On the same folds it reaches 0.599 accuracy and 0.710 respiratory AUC against 0.739 and 0.782, and the proposed model is ahead in all ten folds on every metric (Wilcoxon on fold-means, $p = 0.002$, the floor at this sample size).

That margin should be read as a statement about this corpus. Their model was built for, and reported on, 877 suspected-obstructive-sleep-apnea recordings; three things had to change to bring it here. iSLEEPS has no oronasal thermocouple, so it reads four of the five signals of their fullest configuration; their respiratory output is a three-way decision every second, which this corpus cannot supply and which we replaced with the same per-epoch binary label used throughout; and a single EEG derivation at their 32 Hz is a thin neural input for staging, which is most of why the staging gap is wider than the respiratory one. The row supports the conclusion that a joint architecture tuned on a general sleep-apnea population does not transfer to a stroke cohort of this size, which is the same conclusion the rest of Table 4 reaches for single-task architectures.

A representative held-out night is shown in Figure 3: the predicted hypnogram follows the reference through the night's cycles, and the stage-probability ribbon exposes where the model is confident and where it hesitates.

The margin over the retrained baselines is tested. The folds are fixed across every model in Table 4, so each baseline is scored on exactly the patients the proposed model was and the comparison is paired. That matters here: fold-to-fold variation is large and it is *shared*, because the difficulty lies in the patients (AttnSleep spans 0.627 to 0.755 across folds). Since that variability is common to models evaluated on identical patient partitions, paired comparisons are more informative than the overlap between marginal standard-deviation ranges. Paired on the ten fold-means, with Holm correction across the eight baselines we ran ourselves, the proposed model is ahead of every one of them on accuracy, by +0.019 to +0.093, and wins between eight and ten of the ten folds in each comparison. The margin is statistically separable for five: Micro SleepNet, SleepTransformer, U-Time, AT-BiLSTM and TinySleepNet (Holm p from 0.016 to 0.039). For the three closest it is not. AttnSleep (+0.023, Holm $p = 0.17$), our reimplementations of the corpus LSTM (+0.019, $p = 0.17$) and 1D-ResNet-SE-LSTM (+0.032, $p = 0.15$) fall within the resolution of ten folds. This is consistent with the cohort-difficulty argument made elsewhere: three architectures spanning 2021 to 2023 land close enough to the proposed model that this cohort does not separate them. The accuracy margin is the narrowest of the three and the one most exposed to the cohort's noise. Macro-F1 and κ show separations two to three times larger, since they reduce the influence of performance on the two majority classes.

F. PER-CLASS STAGING PERFORMANCE

Table 5 breaks staging down by stage. The pattern is the one this cohort imposes on every method: N2, Wake, and

TABLE 5. Per-class staging F1 (Wake, N1, N2, N3, REM), ten-fold patient-independent, at the hidden-Markov operating point. The neural-only and full rows are means over the same thirty fold-values. Best per column in bold.

Model	W↑	N1↑	N2↑	N3↑	R↑
Raw multimodal CNN	0.74	0.37	0.73	0.56	0.67
MM-Net (neural only)	0.79	0.28	0.80	0.71	0.76
MM-Net (neural + cardio)	0.79	0.30	0.80	0.71	0.75

REM are recovered well, N3 is moderate, and N1 is the hard stage, because it is transient and rare. The raw convolutional network reaches the highest N1 F1, which fits its sensitivity to short events, while the proposed model leads on the stable stages. The row-normalised confusion matrix (Fig. 4) shows where the errors land: N1 is absorbed mainly into Wake and N2, and part of N3 is read as N2. Comparing the neural-only and full rows isolates what the cardiorespiratory stream contributes to staging, and the answer is that it contributes nothing measurable. Overall accuracy is 0.741 without the stream and 0.739 with it; per stage, N1 improves by 0.019 (Wilcoxon $p = 0.11$ on ten fold-means) and no stage changes significantly ($p \geq 0.15$; REM shifts by 0.01 in Table 5, well inside that).

This is consistent with the physiology. Sleep stages are scored from cortical, ocular and muscular activity, none of which the cardiorespiratory channels observe, so the absence of a staging gain is the expected result. The clinically relevant quantity is the pair taken together: the second stream yields respiratory-event detection at 0.782 AUC while staging is unchanged, so the second output is obtained without a reduction in the first. The modality-ablation grid of Section VI-G shows the same separation by the complementary experiment, each stream supporting the output its physiology is closest to. That the small staging gain lands on N1 is also consistent with its physiology: it is the transitional stage, the hardest to score, and the one most often coincident with a respiratory event.

G. MODALITY-ABLATION GRID

This ablation is also our attribution method. Because the model is feature-based and never sees a raw spectrogram or a raw EEG trace, a gradient heat-map or Grad-CAM would have no faithful spatial support on the neural side, which is where a staging attribution would be read; we therefore attribute each output by *interventional* ablation, deleting a physiological channel and measuring the change. This is a falsifiable test of what each modality contributes. The proposed model stages sleep at 0.739 accuracy (κ 0.634) and detects respiratory events at 0.782 AUC (0.419 average precision). To test the paper's central claim, that each modality carries the task its physiology is closest to, we remove each modality in turn and keep the rest, on the same folds, with three seeds giving thirty fold-values per row (Table 7). The results show a task-specific separation between streams. Within the cardiorespiratory stream the pattern is not the one the engineered features gave.

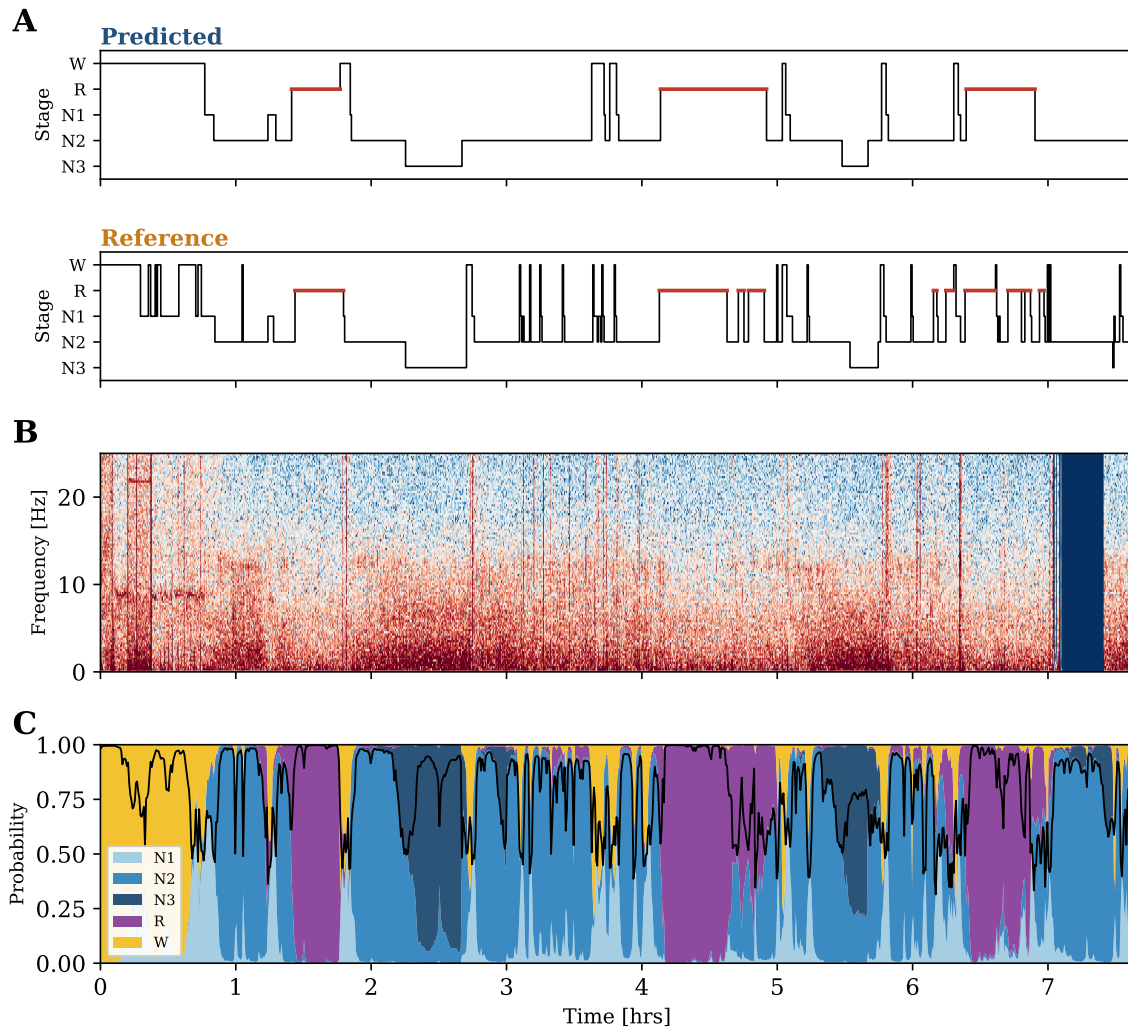


FIGURE 3. A representative held-out night (patient SN90, staging accuracy 0.80). (A) Predicted and reference hypnograms, REM highlighted. (B) EEG spectrogram, C4 derivation, 0–25 Hz. (C) Per-epoch stage-probability ribbon; the black line is the maximum posterior.

Removing the whole cardiorespiratory stream drops respiratory AUC from 0.782 to 0.654, the largest single effect in this study, and leaves staging untouched (0.741, n.s.). Removing the neural stream does the reverse: staging falls from 0.739 to 0.680 while respiratory detection holds. No neural ablation moves respiratory AUC and no cardiorespiratory ablation moves staging, so each stream serves the outcome its physiology is closest to.

Within the cardiorespiratory stream, however, only respiratory effort is individually necessary (-0.038 AUC, Holm $p = 0.018$). Removing SpO_2 , pulse/HRV, ECG or airflow individually costs at most 0.005 AUC and none survives correction; SpO_2 , which the engineered-feature configuration was built around, is here among the least necessary channels (-0.002 , $p = 1.00$). The five individual removals sum to -0.046 AUC against -0.128 for removing the stream as a whole, a factor of 2.8. The learned encoder therefore distributes respiratory information redundantly across channels: each is individually dispensable because the others overlap it, while the ensemble

is essential. That redundancy is clinically favourable, since twelve of the ninety-nine recordings are missing at least one cardiorespiratory channel, and it is physiologically coherent: the effort belts register the paradoxical thoraco-abdominal motion of an obstructive event as it happens, whereas desaturation is a lagged consequence of it.

H. WHAT THE JOINT OBJECTIVE CONTRIBUTES

The ablation grid establishes which physiology carries which output. It does not establish whether the two outputs should share a network at all: a model producing two outputs demonstrates integration, and integration alone does not demonstrate benefit. The control is a pair of single-task models, and it has to be matched properly or it measures the wrong thing. Ours changes one line. The architecture, the ten folds, the three seeds, the optimiser, the budget and the cardiorespiratory bypass are those of the full model; only the surviving term of $\mathcal{L} = \mathcal{L}^{\text{stg}} + \lambda \mathcal{L}^{\text{apn}}$ differs, so the staging-only model still builds the cardiorespiratory stream and still feeds the respi-

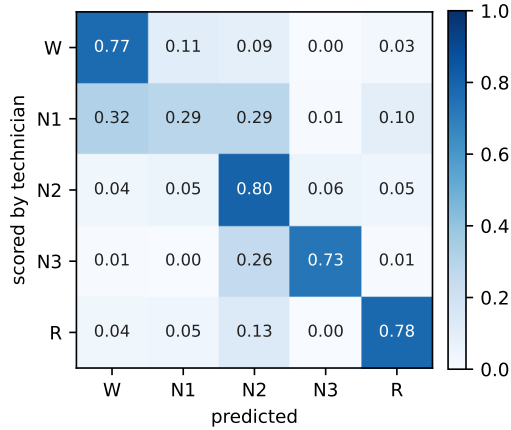


FIGURE 4. Row-normalised confusion matrix for the proposed model (pooled ten-fold test epochs, hidden-Markov operating point). Errors concentrate on N1, which is absorbed into Wake and N2.

ratory head and simply never receives a respiratory gradient. What separates the rows is the objective.

Training both tasks together is better than training either alone (Table 6). Staging rises from 0.733 to 0.739 accuracy, in nine folds of ten (Wilcoxon on the ten fold-means, $p = 0.014$), and κ from 0.625 to 0.634 on the same nine. Respiratory AUC rises from 0.774 to 0.782 in seven folds of ten ($p = 0.049$). Average precision moves +0.005 and does not separate ($p = 0.63$), which we report as the null it is.

Two readings of that are available and only the narrower one is supported. The gains are small: 0.006 accuracy against a fold-to-fold SD of 0.019, and 0.007 AUC against 0.038, so roughly a third and a fifth of the spread the cohort already imposes. What carries them is consistency, nine and seven folds of ten moving the same way, which is what the paired test is reading. The claim we make is therefore that the second task costs the first nothing and returns a little to it, which is the claim the architecture needs: a clinician gets the respiratory read-out without trading away staging accuracy for it. We do not claim multi-task learning as a source of large gains here.

The two diagnostic cells are what make the comparison trustworthy. The respiratory-only model stages at 0.303 accuracy and κ 0.039, and the staging-only model detects at 0.495 AUC. Both are chance for their untrained head, which confirms that the switch isolated the loss and did not quietly leave the other objective running.

I. RECORDINGS WITH AN INCOMPLETE MONTAGE

Those twelve recordings are scored with the absent channels zero-filled instead of being excluded, which raises a fair question: does the network read missingness instead of physiology? It does not appear to. Removing all twelve from the pooled respiratory evaluation moves AUC from 0.778 to 0.775 and moves average precision the other way, from 0.419 to 0.439, so the respiratory result does not rest on them. Predicted per-patient burden is independent of whether a montage was complete ($p = 0.91$), so the shortcut is not

TABLE 6. The joint objective against matched single-task models, ten-fold patient-independent, three seeds. Every row shares the architecture, folds, seeds and budget; only the surviving loss term differs. p is Wilcoxon on the ten fold-means, the unit used throughout. The upper cells of each block are the metric that model was trained for; the lower, in grey, are its untrained head and are chance by construction.

Trained on	Metric	Single	Joint	p
Staging only	Accuracy	0.733	0.739	0.014
	κ	0.625	0.634	0.014
Respiratory only	AUC	0.774	0.782	0.049
	AP	0.414	0.419	0.63
<i>Untrained heads, reported as the check that the switch worked</i>				
Respiratory only	Accuracy	0.303	—	—
Staging only	AUC	0.495	—	—

available even in principle.

Staging is lower on those twelve, 0.670 against 0.743 ($p = 0.017$), which calls for a control. The neural-only configuration receives no cardiorespiratory channel at all and so has nothing zero-filled, yet it shows the same gap in the same direction, 0.689 against 0.744 ($p = 0.040$). Most of the difference therefore belongs to the recordings, which are harder for a model that cannot see the missing channels either. On those twelve the full model sits 0.019 below neural-only, which at $n = 12$ is not distinguishable from noise (Wilcoxon $p = 0.18$), so a small cost can be bounded but not excluded. On the eighty-seven complete recordings the two agree to within 0.002. Attention and plain concatenation stage and detect alike, so the gain comes from the added modality, and the fusion mechanism has little effect.

J. PHYSIOLOGICAL VALIDITY OF THE RESPIRATORY READ-OUT

Because the events were scored from airflow, predominant reliance on that channel could produce a circular association. Removing the airflow features leaves detection essentially unchanged, at 0.778 AUC (Table 7), within one fold standard deviation of the full 0.782, so the model does not appear to depend primarily on the signal used for event annotation. Respiratory effort provides the strongest contribution, being the only channel whose removal costs significantly, together with oxygen desaturation, beat-to-beat cardiac variability, and the cortical arousal that ends the event, none of which is individually indispensable because the learned encoder represents them redundantly. This indicates that detection is supported by physiological correlates of disordered breathing rather than by reproduction of the primary scoring signal. It is also the physiologically expected one: the effort belts register the paradoxical thoraco-abdominal motion of an obstructive event while it is happening, whereas desaturation follows it by several breaths.

TABLE 7. Modality-ablation grid (leave-one-out), ten-fold patient-independent, three seeds. Each row removes one modality and keeps the rest; cells are means over thirty fold-values. p is Wilcoxon on the ten fold-means, Holm-corrected across the nine ablations within each metric and bold where it clears 0.05; the floor at this sample size is 0.018.

Modality removed	Staging acc $\uparrow$	Holm p	Respiratory AUC $\uparrow$	Holm p
none (full model)	0.739 $\pm$ 0.020	—	0.782 $\pm$ 0.038	—
EEG (neural)	0.680	0.018	0.774	0.92
EOG (ocular)	0.731	0.08	0.777	1.00
EMG (muscle)	0.735	1.00	0.778	1.00
SpO ₂	0.742	1.00	0.780	1.00
respiratory effort	0.741	1.00	0.744	0.018
pulse / HRV	0.741	1.00	0.785	1.00
ECG	0.737	1.00	0.777	1.00
airflow	0.740	1.00	0.778	1.00
all cardiorespiratory	0.741	1.00	0.654	0.018

TABLE 8. Event-level detection, pooled over the ten patient-independent folds (8,400 scored events over 771 hours). Within a patient an event is a maximal run of consecutive epochs, with no gap tolerance, and counts as detected when any epoch inside it is flagged; a false alarm is a flagged run overlapping no scored epoch, and the rate denominator is the whole recording, wake included. One flagged run spanning several events therefore marks each detected while costing at most one false alarm, which is where the under-counting below comes from. The flagged fraction is the share of all epochs the threshold marks.

Threshold	Event sens. $\uparrow$	False alarms/h $\downarrow$	Epochs flagged
0.2	0.854	3.21	59%
0.3	0.792	3.25	49%
0.4	0.727	3.05	41%
0.5	0.660	2.82	34%
0.6	0.576	2.42	27%
0.7	0.469	1.94	20%

Per-patient agreement with the clinical index

Aggregated epoch probability (burden)	$\rho = 0.491$
Counted events per hour	$\rho = 0.148$
Predicted vs scored event rate	4.8 vs 11.0

K. RESPIRATORY BASELINES AND PER-EVENT-TYPE DETECTION

The respiratory task needs real competitors (Table 9), on the same folds. The classical baselines read the 14 engineered cardiorespiratory descriptors; the proposed model reads the raw signal through its learned encoder, and its own score on the 14 descriptors is 0.698 (Table 11), which is the like-for-like comparison. A clinical desaturation rule reaches 0.596 AUC, logistic regression 0.582, and gradient boosting 0.670; a random classifier scores 0.157 average precision, the event prevalence. The proposed model reaches 0.782 AUC and 0.419 average precision, above gradient boosting but by a modest margin, so the contribution is the joint single pass and the physiological attribution above; we do not claim state-of-the-art detection accuracy. Detection also separates by event type: the model scores 0.748 AUC on hypopneas (the mild, dominant 81% of events), 0.898 on obstructive apneas, and 0.935 on central apneas. It is strongest on the most severe events, and the overall AUC is held down by the abundant hypopneas.

Scoring at the event level worsens the picture (Table 8). No

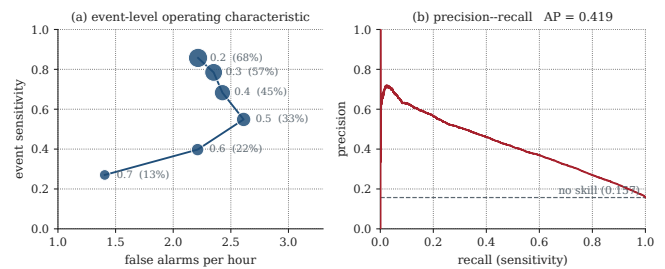


FIGURE 5. Respiratory-event detection, pooled over the ten patient-independent test folds. (a) Event-level operating characteristic. Each point is a decision threshold, labelled and sized by the fraction of all epochs it flags. (b) Precision-recall; the dashed line is the 0.157 event prevalence, the no-skill floor.

threshold is comfortable: the one that finds most events flags three fifths of the night, and the one that flags least finds under half of them. More tellingly, the model recovers the wrong *number* of events, and its per-patient event count agrees with the clinical index far more weakly than its aggregated epoch probability does ($\rho = 0.148$ against $\rho = 0.491$). Consecutive events separated by a few epochs merge into one flagged run, so an epoch-level detector systematically under-counts. This is why we report a burden, and why event-level scoring is the natural next step.

Figure 5 shows the operating characteristics behind these numbers, computed from the pooled held-out predictions of all ten folds. The precision-recall panel is the more informative of the two: respiratory events occupy 16% of epochs, and under that imbalance a ROC curve flatters any detector. Read at fixed sensitivity, the trade-off is explicit: at 0.80 sensitivity the model holds 0.59 specificity and 0.27 precision, and pushing sensitivity to 0.90 costs both, 0.40 and 0.22. Either way this is a research-grade burden estimate, not a scoring-grade detector, and we describe it as such. We also note that pooling every fold's epochs into a single curve is not the same estimator as averaging the ten per-fold scores: pooling weights folds by epoch count and mixes their score scales, giving a pooled AUC of 0.777 against the cross-validated 0.782 $\pm$ 0.038 reported in Table 9. Both are stated so the difference is visible; at 0.005 it is an eighth of one fold standard deviation.

TABLE 9. Respiratory-event detection, ten-fold patient-independent. (a) Detectors on the same folds; the classical baselines read the 14 descriptors, MM-Net the raw signal. Chance average precision is the 0.157 prevalence. (b) AUC by event type, 96 annotated patients. Best per column in bold.

(a) Detection method		
Method	AUC↑	AP↑
Desaturation rule	0.596	0.214
Logistic regression	0.582	0.193
Gradient boosting	0.670	0.290
MM-Net (proposed)	0.782	0.419

(b) MM-Net AUC by event type		
Event type	Positive epochs	AUC↑
Hypopnea	11,605	0.748
Obstructive	1,661	0.898
Central	831	0.935

L. CLINICAL VALIDATION AGAINST AHI AND SEVERITY

Aggregating the per-epoch respiratory probability into a per-patient burden and correlating it with the clinical apnea-hypopnea index (AHI) from the metadata gives a significant positive association (Spearman $\rho = 0.491$, $p < 0.001$, $n = 99$, the twelve recordings with an incomplete cardiorespiratory montage scored with those channels zero-filled; Fig. 6), so the model's output tracks a clinician's summary measure beyond an epoch-level score. Staging, in turn, degrades as sleep-disordered breathing worsens: accuracy falls from 0.769 in patients with a normal AHI to 0.719 in the severe group (Table 10). The pathology that motivates the second output also makes the first harder, which is the clinical reality of this cohort.

The association is statistically significant but modest in magnitude. Three mechanisms plausibly limit it, and each is a property of the formulation rather than of the model. First, mean per-epoch probability is a poor estimator of an hourly event rate: a 30-second epoch holding one hypopnea and one holding three both contribute a single positive, so the aggregate compresses exactly the variation the index is built from. Second, the clinical index is normalised by total sleep time, whereas our burden averages over every scored epoch including wake, and the two denominators diverge most in the patients with the most fragmented sleep, which in this cohort is the severe group. Third, the AHI itself carries substantial night-to-night variability, so a single-night reference sets a ceiling on any correlation with it. We therefore read $\rho = 0.491$ as evidence that the read-out tracks severity at the group level, without claiming that it estimates an index.

M. PERMUTATION IMPORTANCE: A SECOND, INDEPENDENT ATTRIBUTION

The ablation grid establishes what the model *needs*: remove a modality, retrain, and observe which head suffers. A complementary question is what a *trained* model actually uses. We therefore permute each modality's features across epochs at

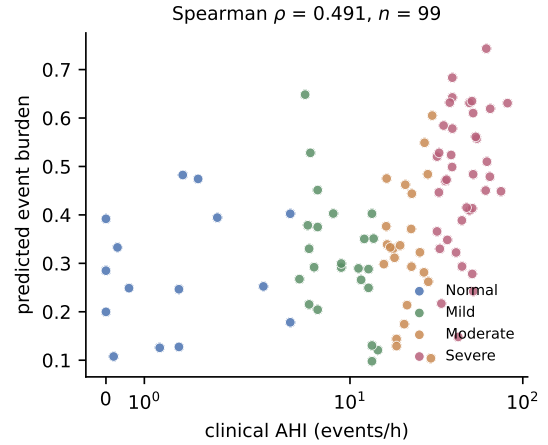


FIGURE 6. Predicted per-patient event burden (mean respiratory probability) against the clinical apnea-hypopnea index, coloured by AASM severity band. The positive association (Spearman $\rho = 0.491$, $n = 99$) shows the model's output tracks a clinician's summary measure.

TABLE 10. Sleep-staging accuracy by sleep-disordered-breathing severity (AASM AHI bands), ten-fold patient-independent; mean $\pm$ SD across patients ($N = 99$). Staging degrades monotonically as severity rises.

AHI band	AHI (events/h)	Patients	Staging acc ↑
Normal	< 5	15	0.769 $\pm$ 0.049
Mild	5–15	23	0.747 $\pm$ 0.069
Moderate	15–30	23	0.722 $\pm$ 0.088
Severe	≥ 30	38	0.719 $\pm$ 0.100

inference time, so the channel is still present but no longer aligned to the epoch it describes, and measure the drop, averaged over three shuffles and the ten fold models.

The two methods can disagree, because a model may need a channel during training yet weight it lightly afterwards. On staging they do not: permuting the neural features costs 0.284 staging accuracy against 0.030 for the ocular and 0.006 for the muscle channel, the same ordering the ablation gives, and the rank correlation between the two attributions is $\rho = 0.833$ ($p = 0.010$).

On the respiratory head the agreement is narrower, and we report the two attributions separately. Both methods place respiratory effort first by a wide margin: 0.038 AUC under retraining, 0.071 under permutation, in each case well clear of the runner-up. The remaining four cardiorespiratory channels, however, have retraining effects that are not distinguishable from zero, so their rank order carries no signal there and the correlation over the five is correspondingly uninformative ($\rho = 0.381$, $p = 0.35$).

Permutation separates them where retraining cannot. Each fold records its own baseline AUC and its AUC after shuffling, so the drop is a within-fold quantity and can be tested directly. Every channel's drop is distinguishable from zero (one-sample on ten per-fold drops, Holm across the eight channels): effort 0.071 ($p < 10^{-4}$), airflow 0.027 ($p = 0.0005$), SpO₂ 0.016 ($p = 0.0002$), pulse/HRV 0.012 ($p =$

0.004), and the remainder below 0.006. Significance here is not importance. The magnitudes span a factor of 32, and what the test establishes is that the *ordering* is real. That ordering is the substantive result: the three channels the model relies on most, effort, airflow and SpO₂, are exactly the three the AASM definition of a respiratory event is written in, and they are one to two orders above the cardiac and muscle channels.

The two experiments answer different questions, so the divergence is informative. Retraining without a channel measures whether it is *irreplaceable*, and the network compensates by re-learning the signal from its correlates; permutation holds the fitted model still and measures what it *uses*. SpO₂ is the clearest case: retraining without it costs 0.002 AUC, shuffling it under the fitted model costs 0.016. A channel can be heavily used and still be replaceable. The claim the comparison supports is therefore the one that matters physiologically: two attributions sharing no machinery, one applied to retrained models and one to fixed ones, independently identify effort as the dominant respiratory cue.

N. DATA EFFICIENCY AND THE COHORT CEILING

Staging on this corpus saturates near 0.72–0.75 for every model family tried here and elsewhere. Whether that reflects the architecture or the cohort is testable: retrain the identical model on a shrinking fraction of the *training patients*, always evaluating on the full held-out fold (Fig. 7). Subsampling is by patient, because epochs within a patient are heavily correlated.

The two heads answer differently. Staging accuracy rises 0.701 → 0.721 → 0.726 → 0.735 as training patients go from 20 to 79, with marginal gains of +0.020, +0.005 and +0.009, all under half a fold standard deviation and flattening. Respiratory AUC over the same range rises 0.732 → 0.750 → 0.752 → 0.785, and its *largest* increment is the final one, +0.033, close to a full fold standard deviation, and steeper than any earlier step. This curve is a single seed on a fixed held-out set, so its endpoint differs slightly from the three-seed headline figures (0.739 and 0.782); the shape is what this curve is here to show. Staging has stopped responding to added patients at this sample size; respiratory detection has not, and is still accelerating. We draw the narrower conclusion the data support: over the range we can test, more patients of this kind would improve breathing detection and would not appreciably improve staging. That is a statement about the returns to cohort size for these methods. It is not a measurement of an irreducible error rate, which fewer than a hundred patients and a finite set of architectures cannot establish. The final model also exceeds the feature-only configuration at the full training-set size, so its advantage is not an artefact of the full sample.

O. HEALTHY-TRAINED MODELS APPLIED TO THE INJURED BRAIN

Table 4 retrains published architectures on iSLEEPS. A different and harder question is what happens when a model built for healthy sleep is applied to this cohort as published.

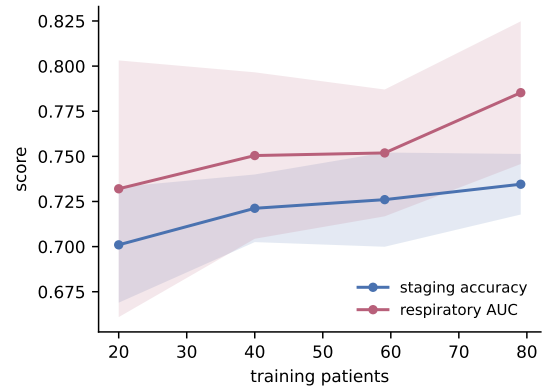


FIGURE 7. Learning curve over training-patient fraction, ten-fold patient-independent, error bars one standard deviation across folds. Staging (blue) flattens by 57 patients; respiratory detection is still climbing at 76. Bands are $\pm$ one standard deviation across folds.

The three architectures are our own: a four-channel CNN, the same CNN with a BiLSTM over epochs, and StagingSeqNet, a deeper convolutional-recurrent stager we built for this comparison. Naming them matters only in that none is a weak straw man; each reaches 0.81 to 0.87 on the healthy corpus it was trained on. We trained them on the Sleep-EDF Cassette subset and applied each to all 99 iSLEEPS patients zero-shot, with no adaptation of any kind. This is the middle block of Table 4, set against the same architectures retrained on this cohort.

The result is not a degradation but a collapse. All three stage healthy sleep competently, 0.806 to 0.855 accuracy and κ 0.73 to 0.81; on stroke the same checkpoints reach 0.306 to 0.312 with κ 0.09 to 0.13, which is close to chance for a five-class problem with this prior. The failure mode is more informative than the headline. Wake takes 62 to 79% of predicted stroke epochs against a true 26.7%, while N2 falls to between 1 and 5% of predictions on the stage that is 42.3% of the data, and N2 recall on the worst of the three is 0.015. These models do not lose accuracy gracefully; they stop recognising sleep architecture at all.

The control is what makes this interpretable. A healthy-trained stager scoring 0.31 on iSLEEPS is evidence of a cohort effect only if the same checkpoint, through the same preprocessing, scores properly on healthy sleep; otherwise a broken input pipeline and a genuine domain gap produce identical numbers. Run through our pipeline on held-out Sleep-EDF recordings, the strongest checkpoint holds at 0.8682 (0.8667 ± 0.0415 per recording), so the pipeline is correct and the collapse belongs to the cohort.

Two bounds, reported together. Sleep-EDF is recorded on Fpz-Cz and Pz-Oz where iSLEEPS uses C4:M1 and O2:M1, so part of this gap is montage shift, not pathology, and quoting the zero-shot number alone would overstate the case. Deep models trained on iSLEEPS with the matched montage reach 0.61–0.69 (upper block), which puts the cost of pathology alone at roughly 0.19 against healthy-sleep performance;

healthy-trained and zero-shot, with the montage shift included, the cost is 0.56. The claim is bounded by those two figures together.

The same argument run in reverse. Every figure above concerns other architectures, which leaves the obvious question unanswered: whether 0.739 is low because the cohort is hard or because the proposed model is weak. So we cross-validated it on healthy sleep, ten-fold and subject-independent over the 21 Sleep-EDF subjects, using the neural branch alone since that corpus carries none of the cardiorespiratory channels we read. It reaches 0.856 ± 0.027 accuracy with κ 0.800 (final row of Table 4). That places it among the healthy-sleep stagers in the middle block, against 0.741 for the identical architecture on stroke. The 0.115 difference is what the cohort costs a model that is demonstrably capable of more, which is a stronger statement than any comparison against other people's architectures can make.

The cohort does not cost every architecture equally. Because the three reimplemented models carry a healthy-sleep score produced by the same code under the same protocol, the table supports a comparison it could not before: how much each model loses in the crossing. TinySleepNet falls 0.148 from healthy sleep to stroke ($0.824 \rightarrow 0.676$), Sleep-Transformer 0.140 ($0.823 \rightarrow 0.683$) and U-Time 0.145 ($0.791 \rightarrow 0.646$), against 0.115 for the proposed model ($0.856 \rightarrow 0.741$). It is therefore not simply higher at both ends; it degrades least. We report this as an observation. Four paired differences will not support a significance test, and the comparison is between systems rather than a controlled ablation, since the proposed rows read engineered features and a frozen embedding over four derivations while the three reimplemented baselines read a single raw channel. The direction is nonetheless the one the rest of the paper predicts: a representation that encodes sleep physiology explicitly survives a disordered cortex better than one a network must discover for itself from fewer than a hundred patients.

Folds are split by subject, not by recording, because Sleep-EDF Cassette records each subject on two nights; splitting on filenames would place the same person on both sides of a fold.

P. STREAM REPRESENTATIONS

The architecture makes two representation choices (engineered features plus a frozen self-supervised embedding on the neural side, a learned convolutional encoder over the raw signal on the cardiorespiratory side), and both were selected by experiment, not assumed. Each family below holds every other hyperparameter fixed and varies one thing, over the same three seeds and ten patient-independent folds, with Holm correction within the family and every test on the ten fold-means.

The engineered cardiorespiratory representation appears to be the principal limitation, and part of the gap can be recovered without additional input information. Each block of Table 11 holds the other stream at its baseline, so no single row there is the full model, and the two baselines differ: the cardiorespiratory block runs the 188 neural descriptors

without the pretrained embedding, while the neural block runs the 14 engineered cardiorespiratory descriptors. That is why the two blocks report different respiratory AUC for what both call an engineered-feature reference (0.698 against 0.685), and why the adopted cardiorespiratory row reads 0.740 staging accuracy against the headline 0.739, which adds the pretrained embedding on top of it. The upper block replaces the 14 engineered cardiorespiratory features with progressively richer representations of the same underlying signal. Every step is significant, and the ordering identifies where the limitation lies. A *random* nonlinear expansion of the same 14 numbers recovers 0.018 AUC, which is obtained without additional input information and with only more capacity to read what was already computed, which says the summary features leave nonlinear structure unrepresented. A random linear projection of the raw waveform adds a further 0.016, so part of what the features discard is recoverable without learning anything. The learned convolutional encoder reaches 0.782, 0.084 above the engineered features.

These results identify the representation of the cardiorespiratory stream, rather than the detector above it, as the factor limiting the respiratory read-out; the detector is held fixed throughout. These results attribute the improvement to the change in representation rather than to the specific encoder architecture.

On the neural side a pretrained encoder does not replace expert features, but does add to them. The lower block of Table 11 varies the neural representation with the cardiorespiratory stream held at the engineered features. Used alone, the frozen encoder is significantly *worse* than the 188 features (-0.029 , Holm $p = 0.008$): pretraining on healthy scalp EEG does not by itself substitute for physiology chosen for this task. Concatenated with the features it adds $+0.008$ accuracy (Holm $p = 0.006$), and a second pretrained encoder on top of the first adds nothing further (-0.004 , $p = 0.56$).

These results support a narrow interpretation: expert physiology and self-supervised pretraining encode partly different information, so their union beats either alone; but the complementarity saturates after one pretrained encoder, so this is not a general argument for stacking foundation models.

Q. THE TWO LOSS CONSTANTS

Two constants in Section V were fixed by argument: the task balance $\lambda = 1$ and the positive weight $\pi = n_-/n_+$, which is 5.35 on this corpus. Equal coefficients do not make equal influence, since the two terms differ in scale and curvature, so $\lambda = 1$ is a claim about arithmetic. We therefore swept each with everything else at the published setting, three seeds by ten folds per value, and evaluate both heads at every point, since the relevant outcome is not a lower value on one head but a trade-off between them.

No trade-off is observed, and the effect of either constant is small. Across λ from 0.25 to 4 staging accuracy spans 0.003 and respiratory AUC 0.018, and across π from 1 to 11 they span 0.003 and 0.007; the fold-to-fold standard deviations are 0.019 and 0.038. Both heads attain their maximum at the same

TABLE 11. What each stream should be represented as. One factor varies per block, everything else held fixed; cells are mean $\pm$ SD over thirty fold-values. * and Holm p are against each block's engineered-feature baseline, the latter on the ten fold-means. "adopted" marks the choice carried into the final model.

Representation	Variant	Resp. AUC $\uparrow$	Staging acc $\uparrow$	Δ	Holm p
Cardiorespiratory	14 engineered features	0.698 $\pm$ 0.051	0.736 $\pm$ 0.014	—	—
	random nonlinear expansion of the 14	0.715 $\pm$ 0.047	0.737 $\pm$ 0.018	+0.018	0.012
	random linear projection of raw signal	0.731 $\pm$ 0.044	0.735 $\pm$ 0.021	+0.034	0.012
	learned CNN, 0.155 M (adopted)	0.782 $\pm$ 0.040	0.740 $\pm$ 0.019	+0.084	0.006
Neural	188 engineered features	0.685 $\pm$ 0.041	0.741 $\pm$ 0.016	—	—
	frozen pretrained encoder alone	0.681 $\pm$ 0.033	0.712 $\pm$ 0.033	−0.029	0.008
	features + pretrained (adopted)	0.690 $\pm$ 0.037	0.749 $\pm$ 0.018	+0.008	0.006
	features + two pretrained encoders	0.684 $\pm$ 0.043	0.745 $\pm$ 0.018	+0.004	0.23

λ , which is inconsistent with a compromise setting: were $\lambda = 1$ balancing one task against the other, the two curves would peak on opposite sides of it.

Neither published value is the maximum of those tested. Staging is highest at $\lambda = 2$ (0.7403) and at $\pi = 11$ (0.7422), and respiratory AUC at $\lambda = 2$ (0.7840) and $\pi = 8$ (0.7846). All of these margins are below a fifth of a fold standard deviation, so the sweep does not identify better settings; it establishes that the objective is insensitive to both constants across the range examined, and that fixing them a priori has no measurable cost on this corpus.

R. CALIBRATION AND SELECTIVE PREDICTION

A clinical reader needs to know not only how often the model is right but whether it knows when it is likely to be wrong. Two measurements answer that, both computed on held-out epochs with the raw posteriors before HMM smoothing.

The model is *overconfident*: expected calibration error, over ten equal-width confidence bins weighted by occupancy and ignoring bins holding fewer than fifty epochs, is 0.076, and the gap runs one way in every bin: epochs it scores above 0.9 are correct 88% of the time. Stated confidences should therefore be read as ordered, not as probabilities, which is the displacement Section V derives arriving in the measurement.

Ranking, however, is what abstention needs. Applying split-conformal prediction with the fold's validation patients as the calibration set gives empirical coverage that tracks the target closely (0.894 at a 0.90 target, 0.943 at 0.95). At $\alpha = 0.10$ the model returns a single stage for 45.8% of epochs and is correct on 86.9% of those, against 59.1% on the epochs where it returns a larger set. The abstention is therefore selecting the genuinely hard epochs, and it concentrates where a scorer would expect: N1 receives a singleton set only 19% of the time, the lowest of any stage and the one human scorers agree on least.

Two caveats attach to the coverage figures rather than to the abstention behaviour. The calibration patients are the same patients that stopped training, so their labels have already influenced which checkpoint was kept and they are not the independent calibration sample split-conformal theory assumes; the coverage we report is therefore empirical and carries no guarantee. And epochs within a night are strongly dependent, so the exchangeable unit is closer to the patient

than to the epoch, which makes these marginal rates across pooled epochs and not coverage for a new patient or within any one stage. A calibration split disjoint from early stopping would settle both, and we did not run one.

S. EXTERNAL VALIDATION ON TWO INDEPENDENT CORPORA

Ten-fold cross-validation is patient-independent but remains within one corpus, one centre and one acquisition protocol. We therefore trained a single model on all 99 iSLEEPS patients and evaluated it, unchanged, on two public corpora with no dataset-specific tuning: no fine-tuning, no threshold search, no re-fitted model or decision threshold (per-recording feature standardisation is applied, as in training), and hidden-Markov transitions estimated from iSLEEPS alone (Table 12).

The two corpora test different things. **Sleep-EDF** [41] is healthy sleep recorded on a frontal montage (Fpz-Cz, Pz-Oz). Its Cassette recordings carry an oronasal airflow trace at 1 Hz but no ECG, effort belts or oximetry, and no scored respiratory events, so six of our seven cardiorespiratory inputs are absent and the respiratory head cannot be scored there; it tests the staging head across both a cohort shift and a montage shift. **ISRUC-Sleep** [52] is a sleep-disorders cohort recorded on the same central and occipital derivations we use (C4, C3, O2, O1 against the contralateral mastoid), with SpO₂, airflow, effort and ECG, and with obstructive, central and mixed apneas and hypopneas scored, so it tests the staging head *without* a montage confound and provides the only external test of the respiratory head. Because Sleep-EDF carries almost none of those channels, that branch is fed zeros there. Zeroing an input the network trained on is a train/test mismatch, not an ablation, so we also trained a model with the branch zeroed throughout: it reaches 0.774 accuracy against the 0.794 reported. The two are separate procedures and neither bounds the other, so we report both and rest nothing on the higher one. The mismatched run could be gaining from batch-normalisation statistics that never saw zeros, in which case 0.774 is the more defensible number, and a reader who prefers it loses nothing in our argument: both sit above the in-domain 0.739. Both corpora also differ from iSLEEPS in montage, so both transfer figures are lower bounds and not clean estimates. ISRUC records each subject

twice; respiratory-event prevalence is 10.1% on the first night and 2.4% on the second, so we report the two separately; pooling would merge a difference that has nothing to do with the model.

The respiratory read-out transfers. On ISRUC night one the model reaches 0.780 AUC, and it does so handicapped, because ISRUC provides no pulse channel and no separate effort trace, leaving two of the seven cardiorespiratory inputs zero-filled. A detector that had latched onto one centre's equipment would not survive that; this one does. The comparison requires matched estimators, which we state before giving the two numbers. The external figure is pooled over all epochs of the night, because per-recording AUC is unusable on this corpus: event prevalence ranges from 0% to 22% across the sixteen recordings, and a night with a handful of positive epochs returns an estimate whose sampling variance swamps any difference between models, while a night with none leaves AUC undefined. The in-domain headline 0.782 is a ten-fold mean. Pooled in-domain, the like-for-like quantity, is 0.778. The external result therefore sits within 0.003 of the in-domain one on either estimator, and nothing in the claim rests on which is chosen. Night two falls to 0.723. Its event prevalence is much lower (2.4% against 10.1%), which does not by itself depress AUC, since ranking is insensitive to the mix; what the scarcity of positives does is widen the uncertainty on that night's estimate, so we read the two nights as one imprecise result.

Staging moves in opposite directions on the two corpora, for reasons the data identify. On Sleep-EDF the model *gains* 0.084 of κ over its in-domain score. A representation that had memorised corpus-specific artefacts would degrade out of corpus, so the straightforward reading is that the features are physiological and that the ceiling reported throughout this paper is a property of *stroke sleep* instead of the model. The learning curve reaches the same conclusion (Section VI-N) by a different route. On ISRUC staging falls by 0.090 accuracy, and the class mix explains much of it: ISRUC carries 15.6% N1 against iSLEEPS' 10.2% and Sleep-EDF's 6.6%, and N1 is the weakest stage in every cohort we have measured (recall 0.168 on ISRUC, 0.294 on Sleep-EDF, and the lowest per-class F1 in-domain).

A montage explanation the data now rule out. An earlier version of this work observed N3 recall of 0.554 on Sleep-EDF with 44% of N3 epochs read as N2, and attributed it to slow-wave activity being frontally dominant while the model reads central and occipital derivations. That account made a prediction: N3 should recover on ISRUC, which uses our derivations exactly. Under the final model the prediction fails, and it fails in the direction that rules the explanation out. N3 recall is 0.841 on Sleep-EDF, the mismatched frontal montage, with 12.7% of N3 read as N2, and 0.630 on ISRUC, the matching one, with 34.8%. Electrode position is therefore not what governs N3 transfer here. The only component that changed between the two models is the neural representation (Section VI-P), which points instead at how the cortex is encoded, though with one external pass per corpus and eight

TABLE 12. External validation, zero-shot: one model trained on all 99 iSLEEPS patients, no dataset-specific tuning. In-domain figures are ten-fold means. Sleep-EDF carries no scored respiratory events and almost none of our cardiorespiratory channels, so the respiratory head is unscored there; the two ISRUC nights differ in event prevalence and are reported separately.

Corpus	Cohort	Rec.	Acc $\uparrow$	κ $\uparrow$	Resp. AUC $\uparrow$
iSLEEPS (in-domain)	stroke	99	0.739	0.634	0.782
Sleep-EDF	healthy	40	0.794	0.718	—
ISRUC, night 1	sleep-disord.	8	0.649	0.536	0.780
ISRUC, night 2	sleep-disord.	8	0.575	0.426	0.723

TABLE 13. Model size and clinical capability. The proposed model is mid-sized and is the only one that produces both a stage and a respiratory-event label at competitive staging accuracy; the raw multimodal CNN produces both but stages at 0.654.

Model	Input	Params $\downarrow$	Outputs
CNN-ResNet18 [17]	1 EEG	~ 11 M	stage
DeepSleepNet [2]	1 EEG	~ 21 M	stage
Raw multimodal CNN	14 ch	0.95 M	stage + apnea
MM-Net (proposed)	7 ch feat + raw	2.76 M	stage + apnea

ISRUC subjects we can identify the effect without attributing it confidently. The prediction was therefore not supported by the final-model analysis.

Two limitations qualify this analysis. Accuracy across cohorts with different class balance is not directly comparable, which is why κ is reported alongside. And ISRUC contributes only eight subjects. Per-recording accuracy spans 0.458–0.748, so the ISRUC figures in Table 12 are quoted to three decimals for consistency with the rest of the paper and should be read to two; per-recording respiratory AUC spans 0.515–0.848, with the widest uncertainty on the nights carrying fewest scored events.

T. MODEL COMPLEXITY AND CAPABILITY

Table 13 sets the proposed model beside representative alternatives on size and capability. At 2.76 M trainable parameters the model is an order of magnitude smaller than DeepSleepNet (~ 21 M) and the corpus's own CNN-ResNet18 (~ 11 M), and comparable to SleepTransformer, so size is not where it wins. It delivers both a competitive stage and a respiratory read-out from one pass over the recording, and none of the staging-only models, at any size, can supply the second output.

VII. DISCUSSION

The result is a single model that reads both the neural and the cardiorespiratory channels of the polysomnogram and addresses two clinical questions at once, in a population in which both are relevant. The learned per-epoch embedding separates by sleep stage while the respiratory structure remains diffuse, consistent with the higher staging accuracy and the lower respiratory AUC.

Staging accuracy remained below 0.75 across the model families evaluated on this cohort, and the proposed model

reaches 0.739 while adding a respiratory capability the staging models lack. The respiratory read-out does not depend on the labelling channel: performance is retained after removal of the airflow channel, with the remaining attribution concentrated on respiratory effort, desaturation and arousal, the same signs a physician reads at the bedside.

The 0.739 staging figure should be judged on this cohort, where healthy-sleep accuracy does not transfer. Deep architectures developed on healthy cohorts show substantial performance degradation here: DeepSleepNet reaches 0.615, CNN-ResNet18 0.617, and a network pretrained on the healthy Sleep-EDF corpus and applied to this cohort 0.623, each roughly fifteen to twenty points below its healthy accuracy of about 0.85. The injured brain distorts the very micro-events, spindles, slow waves, and atonia, that those models learned to read. Physiologically grounded modelling on engineered features retains accuracy of 0.72 to 0.75, matching the published baseline on this corpus. Taken together, these observations indicate that architectures developed on healthy sleep do not transfer to this population, and that cohort-specific physiologically grounded multimodal modelling is required.

The modality-ablation results demonstrate a task-specific dependence on neural and cardiorespiratory information: the cortical channels support sleep staging and the cardiorespiratory channels support respiratory-event detection. That the neural and multimodal variants stage alike is the expected outcome, since the second modality adds a capability while leaving staging unchanged. This separation is favourable for clinical use, since each output can be attributed to the physiology it derives from.

Modality-specific contributions do not imply that the tasks should be trained independently. Section VI-H trains each task alone and finds the joint model ahead of both, slightly and consistently. This is the behaviour a shared temporal decoder would predict: the streams carry different information, and the recurrent state that reads them benefits from being trained on two objectives over the same ten-minute window.

A. LIMITATIONS

The system is offline, and mildly transductive. Two properties follow from the design and neither is visible in the numbers. Staging uses whole-night Viterbi decoding, so no epoch can be scored until the recording ends, which rules out the real-time use a bedside monitor would need. And features are standardised per patient, which at test time uses that night's own statistics; the model never sees another patient's labels, so this is not label leakage, but it is a transductive step and a strictly causal deployment would need running or population statistics instead. We quantified the effect of this substitution and observed no measurable performance degradation within the resolution of the present evaluation. Repeating the ten folds with the standardisation statistics pooled over each fold's training patients and applied unchanged to the held-out ones moves accuracy by +0.003, κ by +0.004, and respiratory AUC by -0.000, none of which is distinguishable

from zero when paired across folds (accuracy $p = 0.28$, AUC $p = 1.00$). These results indicate that the reported performance is not dependent on per-recording standardisation. Whole-night decoding therefore remains the primary constraint on real-time bedside deployment, and is unaffected by this substitution.

One hypothesis evaluated in the external analysis was not supported. The montage account of N3 transfer described in Section VI-S does not survive retesting under the final model, and we report it as falsified. Two cautions follow for every other per-class figure in that section: each rests on a single external pass with no fold structure, so none carries an interval, and ISRUC contributes eight subjects, at which size a per-class rate is an estimate whose width we cannot quote. The per-class transfer results are therefore interpreted qualitatively rather than as precise estimates of effect magnitude.

The evidence comes from a single center and 99 patients. Section VI-S shows both heads transfer to independent corpora without tuning, but neither external corpus is a stroke cohort, so every stroke-specific claim rests on iSLEEPS alone; multi-center validation on a second stroke cohort is required before clinical use, and no such public corpus currently exists. The external respiratory result rests on eight subjects, two of whose seven cardiorespiratory inputs were unavailable, so it establishes that the read-out transfers, not how well. Respiratory detection, at 0.782 AUC, is a useful signal but not yet a screening-grade one, and it is scored per epoch. At event level it detects 66% of events at 2.8 false alarms per hour and under-counts the event rate by more than half, 4.8 against 11.0 per hour, so it cannot substitute for a scored index. The staging head is also overconfident (expected calibration error 0.076), so its posteriors should be used for ranking and abstention. Staging does not improve on the published baseline for this corpus, and while every architecture we ran saturates in the same narrow band, we have not established that the band is a limit rather than the reach of the methods and the sample size we had. Finally, the model is feature-based: it inherits the assumptions of the engineered representations and is only partly an end-to-end raw-signal learner, since the neural stream reads engineered descriptors and a frozen embedding while the cardiorespiratory stream is trained on the raw signal, a deliberate trade for data efficiency on a small cohort.

B. FUTURE WORK

Future evaluation should include event-level metrics, including event sensitivity and false alarms per hour, which are more directly relevant to clinical scoring than the epoch-level AUC reported here. A further direction is to evaluate whether the multimodal representation is associated with stroke severity and recovery outcomes.

VIII. CONCLUSION

We built a model that reads the neural and cardiorespiratory channels of a stroke patient's polysomnogram and, from one pass, stages their sleep and detects their respiratory events.

Deep networks developed on healthy-sleep cohorts achieved 0.61–0.72 accuracy on iSLEEPS, whereas MM-Net achieved 0.739, the highest of the model families evaluated here. Its principal contribution is the concurrent respiratory read-out at 0.782 AUC, which is retained after removal of the airflow channel and therefore does not depend on the signal the events were scored from. Reading the full recording allows a single model to report on breathing and on sleep in a population in which sleep-disordered breathing is prevalent. Code and features are released.

APPENDIX A THE 188 NEURAL DESCRIPTORS

Each 30-second epoch is 3000 samples at 100 Hz, and every descriptor below is computed on one epoch alone. Nothing is smoothed across epochs, because the cross-epoch context the model needs is supplied by the recurrent decoder rather than by the features. Twenty-three descriptors are computed on each of the seven neural channels, C4:M1, C3:M2, O2:M1, O1:M2, E1:M2, E2:M2 and the chin EMG, which gives 161; a further 27 event descriptors are applied only to the channels whose physiology defines them, giving 188 in total.

The per-channel set begins with the spectrum, a Welch estimate over 256-sample segments overlapping by 128. From it we take absolute and relative power in the five bands of Section V-B, the logarithm of total power, spectral entropy on the normalised spectrum, the 95% spectral-edge frequency and the mean frequency. Six time-domain statistics follow, namely standard deviation, peak-to-peak, root mean square, skewness, kurtosis and zero-crossing rate, and to these are added the three Hjorth descriptors, activity, mobility and complexity, computed from the first and second differences [47].

The event descriptors follow the graphoelements a scorer reads, and every band-pass among them is a fourth-order Butterworth applied forwards and backwards so that phase is preserved. Spindles contribute three descriptors per EEG channel: the channel is filtered to 11–16 Hz, its analytic envelope is taken by Hilbert transform, and the threshold is set at the 90th percentile of that envelope within the epoch, so that it tracks the amplitude of the epoch itself instead of a fixed microvolt level; density counts upward crossings of that threshold, and the mean and standard deviation of the envelope complete the three. Slow waves contribute two per EEG channel, the peak-to-peak and the mean absolute amplitude of the 0.5–4 Hz band, with no duration or morphology criterion imposed, so that they summarise slow-wave amplitude and do not detect discrete waves. Each EOG channel contributes the log mean squared first difference and the 95th percentile of its absolute value, which separate the fast deflections of REM from slow drift, and the chin EMG contributes log root-mean-square, the 90th percentile of the absolute signal, and the log variance of the first difference, which together track tonic level and its loss in REM.

No epoch and no channel is rejected, and no amplitude threshold is applied, so a corrupted channel reaches the model. The modality ablations of Section VI-G show the streams carry enough redundancy to absorb one, and any re-

jection rule would have to be tuned on this cohort. Non-finite values, which arise only from a flat channel, are set to zero. Each descriptor is finally standardised against the mean and standard deviation of its own recording, the transductive step Section VII-A returns to, and nothing is aggregated across epochs.

DATA AND CODE AVAILABILITY

The iSLEEPS corpus is publicly available (Zenodo, DOI 10.5281/zenodo.14873844; and the india-data.org portal) as described by Maiti *et al.* [14]. Our code, engineered features, and reproduction notebook are released at <https://github.com/wshuv-o/isleeps-sleep-staging>.

ETHICS

The iSLEEPS data collection was approved by the NIMHANS Institutional Ethics Committee (No. NIMHANS/34th IEC (BS&NS DIV.)/2022, dated 05.02.2022) [14]. This work is a secondary analysis of the de-identified public release and required no additional enrollment.

FUNDING

The authors received no specific funding for this work.

ACKNOWLEDGMENT

The authors thank the iSLEEPS investigators for releasing the corpus.

REFERENCES

- [1] American Academy of Sleep Medicine, *The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications, Version 3*. American Academy of Sleep Medicine, 2023.
- [2] A. Supratak, H. Dong, C. Wu, and Y. Guo, “DeepSleepNet: A model for automatic sleep stage scoring based on raw single-channel EEG,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 25, no. 11, pp. 1998–2008, 2017. [Online]. Available: <https://doi.org/10.1109/TNSRE.2017.2721116>
- [3] H. Phan, F. Andreotti, N. Cooray, O. Y. Chén, and M. De Vos, “SeqSleepNet: End-to-end hierarchical recurrent neural network for sequence-to-sequence automatic sleep staging,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 27, no. 3, pp. 400–410, 2019. [Online]. Available: <https://doi.org/10.1109/TNSRE.2019.2896659>
- [4] A. Supratak and Y. Guo, “TinySleepNet: An efficient deep learning model for sleep stage scoring based on raw single-channel EEG,” in *42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, 2020, pp. 641–644. [Online]. Available: <https://doi.org/10.1109/EMBC44109.2020.9176741>
- [5] E. Eldele, Z. Chen, C. Liu, M. Wu, C.-K. Kwok, X. Li, and C. Guan, “An attention-based deep learning approach for sleep stage classification with single-channel EEG,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 809–818, 2021. [Online]. Available: <https://doi.org/10.1109/TNSRE.2021.3076234>
- [6] H. Phan, O. Y. Chén, M. C. Tran, P. Koch, A. Mertins, and M. De Vos, “XSleepNet: Multi-view sequential model for automatic sleep staging,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 9, pp. 5903–5915, 2022. [Online]. Available: <https://doi.org/10.1109/TPAMI.2021.3070057>
- [7] M. Perslev, M. H. Jensen, S. Darkner, P. J. Jennum, and C. Igel, “U-Time: A fully convolutional network for time series segmentation applied to sleep staging,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 4417–4428.

- [8] W.-B. Jiang, L.-M. Zhao, and B.-L. Lu, "Large brain model for learning generic representations with tremendous EEG data in BCI," in *International Conference on Learning Representations (ICLR)*, 2024, arXiv:2405.18765.
- [9] J. Wang, S. Zhao, Z. Luo, Y. Zhou, H. Jiang, S. Li, T. Li, and G. Pan, "CBraMod: A criss-cross brain foundation model for EEG decoding," in *International Conference on Learning Representations (ICLR)*, 2025, arXiv:2412.07236.
- [10] G. Wang, W. Liu, Y. He, C. Xu, L. Ma, and H. Li, "EEGPT: Pretrained transformer for universal and reliable representation of EEG signals," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024. [Online]. Available: <https://doi.org/10.52202/079017-1239>
- [11] C. Yang, M. B. Westover, and J. Sun, "BIOT: Biosignal transformer for cross-data learning in the wild," in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023, pp. 78 240–78 260. [Online]. Available: <https://openreview.net/forum?id=c2LZyTyddi>
- [12] H. Phan and K. Mikkelsen, "Automatic sleep staging of EEG signals: Recent development, challenges, and future directions," *Physiological Measurement*, vol. 43, no. 4, 2022. [Online]. Available: <https://doi.org/10.1088/1361-6579/ac6049>
- [13] K. Yang and H. Wang, "Deep learning for EEG-based sleep stage classification: A review," *Medical & Biological Engineering & Computing*, vol. 64, pp. 2017–2043, 2026. [Online]. Available: <https://doi.org/10.1007/s11517-026-03583-3>
- [14] S. Maiti, S. K. Sharma, S. Mythirayee, S. Rajendran, and R. S. Bapi, "Polysomnography dataset for sleep analysis in ischemic stroke patients," *Scientific Data*, vol. 13, no. 1, p. 421, 2026. [Online]. Available: <https://doi.org/10.1038/s41597-026-06747-w>
- [15] C. L. Bassetti and M. S. Aldrich, "Sleep electroencephalogram changes in acute hemispheric stroke," *Sleep Medicine*, vol. 2, no. 3, pp. 185–194, 2001. [Online]. Available: [https://doi.org/10.1016/s1389-9457\(00\)00071-x](https://doi.org/10.1016/s1389-9457(00)00071-x)
- [16] C. Denis, I. Jausse, L. Guiraud, C. Mestejanot, C. Arquizan, I. Mourand, S. Chenini, B. Abrial, A. Wacongne, R. Tamisier, S. Baillieu, J.-L. Pepin, L. Barateau, and Y. Dauvilliers, "Functional recovery after ischemic stroke: Impact of different sleep health parameters," *Journal of Sleep Research*, vol. 33, no. 1, 2024. [Online]. Available: <https://doi.org/10.1111/jsr.13964>
- [17] S. Bose, S. Maiti, S. K. Sharma, S. Mythirayee, T. Chakraborti, S. Rajendran, and R. S. Bapi, "AI generalisation gap in comorbid sleep disorder staging," in *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2026. [Online]. Available: <https://doi.org/10.1109/ISBI61048.2026.11515484>
- [18] R. Huttunen, T. Leppänen, B. Duce, E. S. Arnardottir, S. Nikkonen, S. Myllymaa, and J. Töyräs, "A comparison of signal combinations for deep learning-based simultaneous sleep staging and respiratory event detection," *IEEE Transactions on Biomedical Engineering*, vol. 70, no. 5, pp. 1704–1714, 2023. [Online]. Available: <https://doi.org/10.1109/TBME.2022.3225268>
- [19] H. Phan, K. Mikkelsen, O. Y. Chén, P. Koch, A. Mertins, and M. De Vos, "SleepTransformer: Automatic sleep staging with interpretability and uncertainty quantification," *IEEE Transactions on Biomedical Engineering*, vol. 69, no. 8, pp. 2456–2467, 2022. [Online]. Available: <https://doi.org/10.1109/TBME.2022.3147187>
- [20] H. Phan, K. P. Lorenzen, E. Heremans, O. Y. Chén, M. C. Tran, P. Koch, A. Mertins, M. Baumert *et al.*, "L-SeqSleepNet: Whole-cycle long sequence modelling for automatic sleep staging," *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 10, pp. 4748–4757, 2023. [Online]. Available: <https://doi.org/10.1109/JBHI.2023.3303197>
- [21] N. Jirakittayakorn, Y. Wongsawat, and S. Mitirattanakul, "ZleepAnlyst-Net: A novel deep learning model for automatic sleep stage scoring based on single-channel raw EEG data using separating training," *Scientific Reports*, vol. 14, no. 1, p. 9859, 2024. [Online]. Available: <https://doi.org/10.1038/s41598-024-60796-y>
- [22] Y. Guo, M. Nowakowski, and W. Dai, "FlexSleepTransformer: A transformer-based sleep staging model with flexible input channel configurations," *Scientific Reports*, vol. 14, no. 1, p. 26312, 2024. [Online]. Available: <https://doi.org/10.1038/s41598-024-76197-0>
- [23] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," *arXiv preprint arXiv:2312.00752*, 2023.
- [24] X. Zhou, Y. Han, Z. Chen, C. Liu, Y. Ding, Z. Jia, and Y. Liu, "BiT-MamSleep: Bidirectional temporal mamba for EEG sleep staging," *arXiv preprint arXiv:2411.01589*, 2024. [Online]. Available: <https://doi.org/10.48550/ARXIV.2411.01589>
- [25] M. Perslev, S. Darkner, L. Kempfner, M. Nikolic, P. J. Jennum, and C. Igel, "U-Sleep: Resilient high-frequency sleep staging," *npj Digital Medicine*, vol. 4, no. 1, p. 72, 2021.
- [26] "A multi-task deep learning approach for simultaneous sleep staging and apnea detection for elderly people," *Interdisciplinary Sciences: Computational Life Sciences*, 2025. [Online]. Available: <https://doi.org/10.1007/s12539-025-00721-7>
- [27] R. Thapa, M. R. Kjaer, B. He, I. Covert, H. Moore IV, U. Hanif *et al.*, "A multimodal sleep foundation model for disease prediction," *Nature Medicine*, vol. 32, no. 2, pp. 752–762, 2026. [Online]. Available: <https://doi.org/10.1038/s41591-025-04133-4>
- [28] Y. Zhou, J. Wu, Z. Ren, Z. Yao, W. Lu, K. Peng, Q. Zheng, C. Song *et al.*, "CSBrain: A cross-scale spatiotemporal brain foundation model for EEG decoding," *arXiv preprint arXiv:2506.23075*, 2025. [Online]. Available: <https://doi.org/10.52202/085713-2915>
- [29] K. Kontras, T. Osselaer, S. G. Mouslech, A.-I. Karaiskou, G. Gagliardi, M. H. Strypsteen *et al.*, "NeuroAtlas: Benchmarking foundation models for clinical EEG and brain-computer interfaces," *arXiv preprint arXiv:2605.14698*, 2026.
- [30] A. Dei Rossi, M. Metaldi, M. Bechny, I. Filchenko, J. van der Meer, M. H. Schmidt, C. L. A. Bassetti, A. Tzovara *et al.*, "SLEEPYLAND: Trust begins with fair evaluation of automatic sleep staging models," *npj Digital Medicine*, vol. 9, no. 1, p. 55, 2026. [Online]. Available: <https://doi.org/10.1038/s41746-025-02237-2>
- [31] E. Estevan, M. Sierra-Torralba, E. López-Larraz, and L. Montesano, "A systematic evaluation of self-supervised learning for label-efficient sleep staging with wearable EEG," *arXiv preprint arXiv:2510.07960*, 2025. [Online]. Available: <https://doi.org/10.48550/ARXIV.2510.07960>
- [32] A. Guillot and V. Thorey, "RobustSleepNet: Transfer learning for automated sleep staging at scale," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 29, pp. 1441–1451, 2021. [Online]. Available: <https://doi.org/10.1109/TNSRE.2021.3098968>
- [33] U. Hanif, A. Aloulou, F. Crosbie, P. Bouchequet, M. Chennaoui, T. Andrilion, and D. Leger, "Deciphering insomnia: Benchmarking automated sleep staging algorithms for complex sleep disorders," *Journal of Sleep Research*, vol. 34, no. 6, p. e70048, 2025. [Online]. Available: <https://doi.org/10.1111/jsr.70048>
- [34] N. Tellenbach, M. H. Schmidt, F. Alexiev, E. Blondiaux, F. Cavalloni, C. L. Bassetti, L. Heydrich, and P. Bargiotas, "REM sleep and muscle atonia in brainstem stroke: A quantitative polysomnographic and lesion analysis study," *Journal of Sleep Research*, vol. 32, no. 2, 2023.
- [35] J. M. Gottselig, C. L. Bassetti, and P. Achermann, "Power and coherence of sleep spindle frequency activity following hemispheric stroke," *Brain*, vol. 125, no. 2, pp. 373–383, 2002. [Online]. Available: <https://doi.org/10.1093/brain/awf021>
- [36] L. M. J. Fernández and A. Lüthi, "Sleep spindles: Mechanisms and functions," *Physiological Reviews*, vol. 100, no. 2, pp. 805–868, 2020. [Online]. Available: <https://doi.org/10.1152/physrev.00042.2018>
- [37] B. K. Simpson, R. Rangwani, A. Abbasi, J. M. Chung, C. M. Reed, and T. Gulati, "Disturbed laterality of non-rapid eye movement sleep oscillations in post-stroke human sleep: A pilot study," *Frontiers in Neurology*, vol. 14, p. 1243575, 2023. [Online]. Available: <https://doi.org/10.3389/fneur.2023.1243575>
- [38] L. Facchin, C. Schöne, A. Mensen *et al.*, "Slow waves promote sleep-dependent plasticity and functional recovery after stroke," *Journal of Neuroscience*, vol. 40, no. 45, pp. 8637–8651, 2020. [Online]. Available: <https://doi.org/10.1523/jneurosci.0373-20.2020>
- [39] Y. Zheng, T. Jing, L. Xu, S. Sun, W. Yu, R. Liu, G. Xu, and C. Chu, "Analysis of the correlation between sleep spindles and cognitive impairment in patients with ischemic stroke," *Scientific Reports*, vol. 15, no. 1, p. 36259, 2025. [Online]. Available: <https://doi.org/10.1038/s41598-025-20102-w>
- [40] A. Agius Anastasi *et al.*, "Brain symmetry index in healthy and stroke patients for assessment and prognosis," *Stroke Research and Treatment*, vol. 2017, 2017. [Online]. Available: <https://doi.org/10.1155/2017/8276136>
- [41] B. Kemp, A. H. Zwiderman, B. Tuk, H. A. C. Kamphuisen, and J. J. L. Oberyé, "Analysis of a sleep-dependent neuronal feedback loop: The slow-wave microcontinuity of the EEG," *IEEE Transactions on Biomedical Engineering*, vol. 47, no. 9, pp. 1185–1194, 2000. [Online]. Available: <https://doi.org/10.1109/10.867928>
- [42] A. L. Goldberger, L. A. N. Amaral, L. Glass *et al.*, "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp.

- e215–e220, 2000. [Online]. Available: <https://doi.org/10.1161/01.cir.101.23.e215>
- [43] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794. [Online]. Available: <https://doi.org/10.1145/2939672.2939785>
- [44] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems 30 (NIPS)*, 2017.
- [45] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, “CatBoost: Unbiased boosting with categorical features,” in *Advances in Neural Information Processing Systems 31 (NeurIPS)*, 2018. [Online]. Available: <https://doi.org/10.48550/arXiv.1706.09516>
- [46] A. N. Angelopoulos and S. Bates, “Conformal prediction: A gentle introduction,” *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.
- [47] B. Hjorth, “EEG analysis based on time domain properties,” *Electroencephalography and Clinical Neurophysiology*, vol. 29, no. 3, pp. 306–310, 1970. [Online]. Available: [https://doi.org/10.1016/0013-4694\(70\)90143-4](https://doi.org/10.1016/0013-4694(70)90143-4)
- [48] W. Chen, Y. Cai, A. Li, Y. Su, and K. Jiang, “Single-channel sleep EEG classification method based on LSTM and hidden markov model,” *Brain Sciences*, vol. 14, no. 11, p. 1087, 2024. [Online]. Available: <https://doi.org/10.3390/brainsci14111087>
- [49] W. Li and J. Gao, “Automatic sleep staging by a hybrid model based on deep 1D-ResNet-SE and LSTM with single-channel raw EEG signals,” *PeerJ Computer Science*, vol. 9, p. e1561, 2023. [Online]. Available: <https://doi.org/10.7717/peerj-cs.1561>
- [50] G. Liu, G. Wei, S. Sun, D. Mao, J. Zhang, D. Zhao, X. Tian, X. Wang, and N. Chen, “Micro SleepNet: Efficient deep learning model for mobile terminal real-time sleep staging,” *Frontiers in Neuroscience*, vol. 17, p. 1218072, 2023. [Online]. Available: <https://doi.org/10.3389/fnins.2023.1218072>
- [51] M. Fu, Y. Wang, Z. Chen, J. Li, F. Xu, X. Liu, and F. Hou, “Deep learning in automatic sleep staging with a single channel electroencephalography,” *Frontiers in Physiology*, vol. 12, p. 628502, 2021. [Online]. Available: <https://doi.org/10.3389/fphys.2021.628502>
- [52] S. Khalighi, T. Sousa, J. M. Santos, and U. Nunes, “ISRUC-Sleep: A comprehensive public dataset for sleep researchers,” *Computer Methods and Programs in Biomedicine*, vol. 124, pp. 180–192, 2016.



MD WAHIDUZZAMAN SUVA received the B.Sc. degree in computer science and engineering from American International University-Bangladesh (AIUB), Dhaka, Bangladesh, in 2025, where he is currently pursuing the M.Sc. degree in computer science. He is an Applied AI Engineer with Odin, Dhaka, and a Co-Founder of Prodigy AI, a non-profit research group focused on brain-computer interfaces. He is a co-author of an article on equivariant quantum graph neural networks in *IEEE Transactions on Artificial Intelligence*, of an article on equivariant quantum clustering in *Physica Scripta*, and of a Springer book chapter on federated learning in healthcare technology. His work spans quantum machine learning, federated learning, and EEG-based brain-computer interfaces. He serves as a reviewer for several international journals and conferences. His research interests include brain-computer and brain-machine interfaces, brain signal analysis, transformer models for EEG, and applied machine learning for health.



ESM-E MOULA CHOWDHURY ABHA received the B.Sc. degree in computer science and engineering from American International University-Bangladesh (AIUB), Dhaka, Bangladesh, in 2025, where she is currently pursuing the M.Sc. degree in computer science. She is with the Department of Computer Science and Engineering, AIUB, and a Researcher with Prodigy AI, a non-profit research group focused on brain-computer interfaces. She has over three years of industry experience as a Backend and AI Application Developer, building scalable API architectures and AI-integrated platforms for fintech, education, and customer-support products. She is the lead author of a Springer book chapter on federated learning in healthcare technology and a co-author of an article on equivariant quantum clustering in *Physica Scripta*, and she has contributed to research on attention-based EEG classification and multimodal deep learning for autonomous vehicle perception. Her research interests include brain-computer and brain-machine interfaces, brain signal analysis, transformer models for EEG, and applied machine learning for health.



MD IMTIAJ ALAM SAJIN received the B.Sc. degree in computer science and engineering, with a major in data science, from American International University-Bangladesh (AIUB), Dhaka, Bangladesh, in 2026, where he is currently pursuing the M.Sc. degree in computer science. He is with the Department of Computer Science and Engineering, AIUB, and a Researcher with Prodigy AI, a non-profit research group focused on brain-computer interfaces. He has also worked as a Software Developer with Byteverse Ltd., Dhaka, contributing to Chatsby, an AI-driven customer-support platform. His B.Sc. thesis proposed a dual-attention transformer for EEG classification in brain-computer interfaces. He is a co-author of an article on symmetry-aware equivariant quantum clustering in *Physica Scripta* and of a Springer book chapter on federated learning in healthcare technology. His research interests include brain-computer and brain-machine interfaces, brain signal analysis, transformer models for EEG, and applied machine learning for health.



MD IFTEKHARUL MOBIN (Member, IEEE) is an Assistant Professor with the Faculty of Science and Technology, American International University-Bangladesh (AIUB), Dhaka, Bangladesh. He received the Ph.D. degree from Queen Mary University of London, London, U.K., and the B.Sc. degree from the Islamic University of Technology (IUT), Gazipur, Bangladesh. He was awarded the Queen Mary scholarship and the ImpactQM scholarship in acknowledgment of his contributions to industry-academia partnerships, and his research has received support from Bangladesh's Ministry of Information and Communication Technology. He served as a software engineer in the United Kingdom, Japan, and Bangladesh for several years, and that dual exposure to academic and industry settings informs his perspective on industry-academia collaboration. He has served on the program committees of international conferences and as an editor and reviewer for conferences and journals worldwide. His research interests include natural language processing, time-series prediction, health informatics, the Internet of Things, and image processing.

...